thiLLMo: Ontology-Grounded RAG for Culturally Faithful Kikuyu-English Proverb Translation

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Introduction

About the Name

thiLLMo is a portmanteau combining:

- **"Thimo"** (pronounced "thee-mo")—the Kikuyu word for proverbs
- **"LLM"**—Large Language Model

Pronunciation: /ilmo/ ("theel-mo")

This name reflects the project's core mission: bridging traditional Kikuyu wisdom (*thimo*) with modern AI technology (LLM) to create culturally faithful translations that preserve the deep cultural significance of traditional sayings.

1.1 Background and Motivation

1.1.1 The Cultural Significance of Proverbs

Consider the Kikuyu proverb *"Mũtumia mũgima ndagagĩrwo nĩ thĩĩna"* ("A good wife is never troubled by poverty"). This eight-word statement encodes complex cultural attitudes about gender roles, economic partnership, and the nature of prosperity itself—knowledge that resists direct translation Finnegan [2012]. Proverbs function as compact repositories of worldview, values, and historical experience, shaping community identities through memorable, transmissible forms. In African oral traditions, they serve as pedagogical tools, vehicles for moral instruction, and mechanisms for social cohesion, embedding complex philosophical and ethical frameworks Mbiti [1990].

For the Kikuyu people of Kenya, proverbs (*thimo*) represent a sophisticated system of knowledge transmission spanning agriculture, kinship, governance, ethics, and economic philosophy. The proverb *"Andu ni indo"* ("People are wealth") exemplifies this depth—it is not merely a statement about valuing relationships, but an entire economic and social philosophy that prioritizes community capital over material accumulation. This single proverb encodes centuries of Kikuyu cultural wisdom about sustainable prosperity, reciprocity systems (*ngwatio*), and collective well-being.

The translation of such proverbs presents unique challenges that extend far beyond direct word-for-word mapping. Proverbs are intricately woven into their cultural contexts, commonly employing figurative language, metaphors, and specific cultural references that lack direct lexical equivalents in other languages Mieder [2004]. A translation that achieves lexical accuracy while losing cultural authenticity fails to preserve the very essence that makes proverbs valuable as cultural artifacts.

1.1.2 The Low-Resource Language Challenge

Low-resource languages (LRLs) like Kikuyu face a severe deficit of high-quality digital resources, parallel corpora, and well-annotated linguistic data Joshi et al. [2020]. Kikuyu, spoken by approximately 7 million people, possesses a rich oral literary tradition but minimal representation in digital language technologies. Existing digital resources, such as Wikipedia entries and online dictionaries, are frequently outdated, incomplete, or culturally decontextualized. This data paucity severely hinders the development of robust machine translation systems capable of handling the subtle and nuanced cultural expressions embedded within proverbs.

The combination of cultural specificity and data scarcity creates a *double-bind problem* for computational approaches. On one hand, neural machine translation models trained on generic corpora cannot interpret the metaphorical meanings, contextual appropriateness, and ethical implications embedded in proverbs—what we might call *cultural opacity*. On the other hand, traditional NMT approaches demand massive parallel corpora (millions of sentence pairs) that simply do not exist for Kikuyu-English proverb translation—a fundamental problem of *data insufficiency*. Neither challenge can be addressed without confronting the other.

This double-bind necessitates architectural innovations that can leverage *structured cultural knowledge* to compensate for *unstructured data scarcity*. Rather than attempting to learn cultural patterns implicitly from large datasets (which are unavailable), we propose explicitly encoding cultural knowledge into a formal ontology that can guide translation processes.

1.1.3 The Promise and Peril of Large Language Models

Recent advances in Large Language Models (LLMs) such as GPT-4, Gemini, and multilingual variants like NLLB-200 Costa-jussà et al. [2022] have demonstrated remarkable capabilities in cross-lingual tasks, including zero-shot and few-shot translation for low-resource languages. These models encode substantial world knowledge from web-scale pretraining, enabling them to generate fluent, grammatically correct translations even for languages with minimal training data.

However, LLMs exhibit critical limitations when confronted with culturally nuanced content:

Hallucination and Fabrication LLMs often generate plausible-sounding but factually incorrect or culturally inappropriate content when operating beyond their knowledge boundaries Zhang et al. [2023]. For proverb translation, this manifests as generating generic English sayings that miss the specific cultural frame of the Kikuyu original.

This problem compounds when we consider the second limitation.

Cultural Homogenization LLM pretraining corpora are dominated by English and other high-resource languages, creating implicit biases toward Western cultural frameworks Blodgett et al. [2020]. When translating Kikuyu proverbs, models may unconsciously map culturally-specific concepts (e.g., *ngwatio* reciprocity systems) onto Western equivalents (e.g., "barter"), losing critical cultural distinctions.

These first two limitations might be tolerable if LLMs provided transparent reasoning, but the third limitation prevents this.

Lack of Structured Knowledge LLMs encode knowledge as probabilistic patterns in continuous vector spaces, making it difficult to verify factual claims, trace reasoning processes, or ensure adherence to cultural constraints Borgeaud et al. [2022]. For tasks requiring precision and cultural fidelity, this "black box" nature is problematic.

These limitations suggest that while LLMs possess powerful generative capabilities, they require *structured augmentation* to achieve genuine cultural faithfulness in specialized domains like proverb translation.

1.2 Problem Statement

The central problem addressed by this thesis is: **How can we achieve culturally faithful translation of Kikuyu proverbs to English when confronted with both data scarcity (inherent to low-resource languages) and the need for deep cultural grounding (beyond the capabilities of current LLMs)?**

This problem manifests across three dimensions:

1.2.1 Technical Challenge: Knowledge Representation

Current Retrieval-Augmented Generation (RAG) systems primarily retrieve unstructured text chunks based on vector similarity Lewis et al. [2020]. While effective for general question-answering, this approach fails to capture the *relational* and *hierarchical* structure of cultural knowledge. A proverb's meaning is not isolated—it connects to broader cultural themes (e.g., *community values*), specific contexts (e.g., *wealth distribution ceremonies*), moral frameworks (e.g., *obligation to elders*), and historical practices (e.g., *traditional banking systems*). These relationships form a semantic network that cannot be adequately represented through unstructured text retrieval.

Research Gap: Existing RAG systems lack mechanisms for ontology-grounded retrieval that preserves the structured, interconnected nature of cultural knowledge when augmenting LLM contexts.

1.2.2 Linguistic Challenge: Metaphor and Idiomaticity

Kikuyu proverbs employ dense metaphorical language drawn from agricultural practices, animal behavior, kinship structures, and historical events specific to Kikuyu culture. For example, *"Aikaragia mbia ta njuu ngigi"* ("He guards his wealth like a stork chasing locusts") references both the stork's precise hunting behavior and cultural associations between vigilance and financial stewardship. Direct translation yields nonsensical English; cultural translation requires understanding the *symbolic function* of the stork metaphor within Kikuyu economic philosophy.

Research Gap: Translation systems lack access to the cultural conceptual mappings required to interpret and convey metaphorical meanings that are opaque outside their originating culture.

1.2.3 Evaluation Challenge: Measuring Cultural Fidelity

Standard automatic metrics for machine translation quality (BLEU, CHRF++, COMET) are demonstrably inadequate for culturally nuanced content Mathur et al. [2020]. These metrics prioritize surface-level lexical overlap and penalize valid paraphrases or cultural adaptations that may be necessary for authentic translation. A translation could score poorly on BLEU while achieving superior cultural fidelity, or vice versa.

Research Gap: Evaluation frameworks for culturally sensitive translation lack robust metrics and methodologies for assessing cultural authenticity alongside linguistic accuracy.

1.3 Formal Hypotheses

To enable rigorous empirical validation of the OG-RAG approach, we formulate testable null and alternative hypotheses grounded in the research gaps identified above. These hypotheses are designed to be falsifiable through statistical testing with significance level $\alpha = 0.05$ and target statistical power of 0.80 Cohen [1988].

1.3.1 Hypothesis 1: Cultural Authenticity

Null Hypothesis ($\mathbf{H}_0^1$): Ontology-grounded RAG does not significantly improve cultural authenticity scores compared to Traditional RAG in Kikuyu proverb translation.

$$H_0^1 : \mu_{\text{OG-RAG}} - \mu_{\text{Trad-RAG}} \leq 0 \tag{1.1}$$

Alternative Hypothesis ($\mathbf{H}_1$): Ontology-grounded RAG significantly improves cultural authenticity scores by at least 10% compared to Traditional RAG.

$$H_1 : \mu_{\text{OG-RAG}} - \mu_{\text{Trad-RAG}} > 0.10 \times \mu_{\text{Trad-RAG}} \tag{1.2}$$

Rationale: The explicit encoding of cultural relationships, metaphorical mappings, and contextual constraints in the ontology should enable more culturally faithful translations than unstructured text retrieval. We predict a medium-to-large effect size (Cohen's $d \geq 0.5$) based on the structured nature of cultural knowledge representation.

1.3.2 Hypothesis 2: Translation Fidelity

Null Hypothesis ($\mathbf{H}_0^2$): Ontology-grounded RAG does not significantly improve translation fidelity scores compared to Traditional RAG.

$$H_0^2 : \mu_{\text{OG-RAG}}^{\text{fidelity}} - \mu_{\text{Trad-RAG}}^{\text{fidelity}} \leq 0 \tag{1.3}$$

Alternative Hypothesis ($\mathbf{H}_2$): Ontology-grounded RAG significantly improves translation fidelity by at least 15% compared to Traditional RAG.

$$H_2 : \mu^{\text{fidelity}}_{\text{OG-RAG}} - \mu^{\text{fidelity}}_{\text{Trad-RAG}} > 0.15 \times \mu^{\text{fidelity}}_{\text{Trad-RAG}} \tag{1.4}$$

Rationale: Structured retrieval should provide more semantically relevant context than similarity-based text matching, enabling the LLM to generate translations that better preserve original meanings. The larger predicted effect size (15% vs. 10%) reflects the hypothesis that ontological grounding benefits semantic precision even more than cultural authenticity.

1.3.3 Hypothesis 3: System Superiority Over Baseline

Null Hypothesis ($\mathbf{H}_0^3$): OG-RAG performance does not significantly exceed raw LLM baseline (zero-shot translation without retrieval augmentation).

$$H_0^3 : \mu^{\text{overall}}_{\text{OG-RAG}} - \mu^{\text{overall}}_{\text{Raw-LLM}} \leq 0 \tag{1.5}$$

Alternative Hypothesis ($\mathbf{H}_3$): OG-RAG achieves at least 5% improvement in overall translation quality compared to the raw LLM baseline.

$$H_3 : \mu^{\text{overall}}_{\text{OG-RAG}} - \mu^{\text{overall}}_{\text{Raw-LLM}} > 0.05 \times \mu^{\text{overall}}_{\text{Raw-LLM}} \tag{1.6}$$

Rationale: This hypothesis validates the fundamental value of retrieval augmentation for culturally nuanced translation. The conservative 5% threshold reflects uncertainty about whether any retrieval augmentation suffices, or whether ontological grounding is specifically necessary. Rejecting H_0^3 while accepting H_1 and H_2 would provide strong evidence for the OG-RAG approach.

1.3.4 Statistical Testing Approach

These hypotheses will be tested using paired t-tests (for within-system comparisons across proverbs) and independent t-tests (for between-system comparisons), with Bonferroni correction for multiple comparisons ($\alpha_{\text{corrected}} = 0.05/3 = 0.0167$). Effect sizes will be reported using Cohen's d for interpretability. Sample size was determined through power analysis to detect medium effects ($d = 0.5$) with 80% power, yielding a minimum requirement of $n = 64$ proverbs per condition; our evaluation uses $n = 100$ for additional robustness Lakens [2013].

The evaluation methodology and detailed results addressing these hypotheses are presented in Chapter 5, with statistical interpretation in Section 5.3.

1.4 Research Objectives and Contributions

This thesis addresses the identified gaps through the following objectives and contributions:

1.4.1 Primary Research Objectives

1. **Develop a formal ontology for Kikuyu proverbs** focused on wealth and prosperity themes, capturing literal meanings, metaphorical interpretations, cultural contexts, usage scenarios, and relationships to broader cultural concepts.

2. **Design and implement an Ontology-Grounded Retrieval-Augmented Generation (OG-RAG) system** that integrates the Kikuyu proverb ontology with state-of-the-art LLMs to enable culturally faithful translation.

3. **Establish a comprehensive evaluation framework** combining expert human assessment with culturally-aware metrics to rigorously measure both linguistic accuracy and cultural fidelity.

4. **Empirically validate** the hypothesis that ontology-grounded retrieval significantly improves cultural authenticity compared to baseline RAG and raw LLM approaches.

1.4.2 Research Contributions

This research makes the following scholarly and practical contributions:

Theoretical Contributions

- **OG-RAG Architecture for Cultural Translation**: A novel application of ontology-grounded RAG specifically designed for culturally nuanced translation tasks, extending prior work on knowledge graph RAG Edge et al. [2024b] to the cultural heritage domain.
- **Cultural Fidelity Framework**: A multi-dimensional evaluation framework operationalizing "cultural fidelity" through measurable components (thematic accuracy, contextual appropriateness, metaphorical preservation, value alignment).
- **Knowledge Representation for Oral Traditions**: Methodological insights into representing tacit, context-dependent knowledge from oral traditions within formal ontological structures.

Practical Contributions

- **Kikuyu Proverb Ontology**: A reusable, machine-readable knowledge base comprising 847 concepts, 1,200+ relationships, and 100+ proverb instances, representing the first formal ontology of Kikuyu cultural knowledge.
- **thiLLMo System**: An open-source implementation of the OG-RAG architecture integrating Neo4j knowledge graphs, vector embeddings, and LLM generation, demonstrating practical feasibility for low-resource cultural translation.
- **Evaluation Benchmark**: A curated dataset of 100 Kikuyu proverbs with expert translations and cultural annotations, enabling reproducible evaluation for future research.
- **Empirical Evidence**: Quantitative demonstration that OG-RAG achieves 10.5% improvement in cultural authenticity and 19.8% improvement in translation fidelity over baseline approaches (Chapter 5).

Broader Impacts

- **Cultural Preservation**: Contributes to digital preservation of Kikuyu intangible cultural heritage by formalizing oral knowledge in accessible, structured formats.
- **Low-Resource Language Technology**: Demonstrates viable architectural patterns for sophisticated NLP applications in low-resource languages without requiring massive training corpora.
- **Ethical AI**: Establishes a model for culturally respectful AI development through collaborative ontology construction with native speakers and transparent knowledge representation.

1.5 Research Questions

The primary research question guiding this work is:

> **RQ-Main**: To what extent can ontology-grounded retrieval augmentation improve the cultural fidelity and linguistic accuracy of LLM-based Kikuyu-to-English proverb translation compared to baseline approaches?

This decomposes into three specific sub-questions addressed empirically in Chapter 5:

RQ1: Quantitative Performance How does the OG-RAG system's translation quality compare to Traditional RAG and Raw LLM baselines across measurable dimensions of cultural authenticity and translation fidelity?

RQ2: Qualitative Mechanisms What qualitative patterns and mechanisms distinguish OG-RAG translations from baseline approaches, and how do these relate to ontological grounding?

RQ3: Failure Analysis Where does OG-RAG still struggle, and what do these limitations reveal about the boundaries of ontology-based cultural knowledge representation?

Answers to these questions (provided in Sections 5.2–5.6) constitute the empirical core of this thesis.

1.6 Scope and Limitations

To maintain research focus and feasibility within a three-month timeframe, this thesis adopts specific scoping decisions:

1.6.1 In Scope

- **Domain**: Kikuyu proverbs specifically addressing themes of wealth, prosperity, and economic wisdom (approximately 100 proverbs from Ireri's collection Ireri [2019b]).
- **Language Pair**: Unidirectional translation from Kikuyu to English.
- **Translation Mode**: Culturally faithful translation (may include paraphrasing, cultural adaptation, or explanatory glosses rather than strict literal translation).
- **Evaluation**: Expert human evaluation by native Kikuyu speakers with cultural competence; LLM-assisted evaluation for supplementary insights.
- **Architecture**: Ontology-grounded RAG combining Neo4j knowledge graphs, semantic embeddings, and GPT-4/Gemini LLMs.

1.6.2 Out of Scope

- **Bidirectional Translation**: English-to-Kikuyu translation is not addressed (requires different cultural adaptation strategies).
- **Comprehensive Corpus**: The ontology covers wealth-related proverbs, not the entirety of Kikuyu proverbial wisdom (estimated 3,000+ proverbs total).
- **Real-time Systems**: System latency optimization and production deployment are deferred to future work.

- **Fine-tuning Approaches**: Due to data scarcity and computational constraints, LLM fine-tuning is not explored; the focus is on prompt-based augmentation.
- **Automatic Metrics Development**: New automatic evaluation metrics are not created; existing metrics are analyzed for limitations.

1.6.3 Known Limitations

- **Ontology Completeness**: The ontology represents current best understanding but cannot capture all tacit cultural knowledge (68% of remaining errors attributed to ontology gaps—see Section 5.5.2).
- **Dialectical Variation**: Kikuyu has three major dialects (Kiambu, Nyeri, Murang'a); the ontology prioritizes standard Kikuyu, potentially missing dialectical nuances (5% of evaluation corpus showed dialect-related ambiguities).
- **Cultural Dynamism**: Proverb meanings evolve with societal changes; the ontology reflects traditional interpretations and may not capture contemporary reinterpretations.
- **Generalizability**: Findings are specific to Kikuyu proverbs and wealth domain; claims about applicability to other LRLs or cultural domains require empirical validation.

These limitations are addressed further in the Discussion (Chapter 6) and suggest directions for future research.

1.7 Thesis Structure

The remainder of this thesis is organized as follows:

Chapter 2: Literature Review surveys the state-of-the-art across three domains: (1) Retrieval-Augmented Generation, from traditional RAG to ontology-grounded approaches; (2) Machine Translation for low-resource languages, with emphasis on cultural challenges; and (3) Knowledge representation for cultural heritage, focusing on ontology engineering methodologies. This chapter establishes the theoretical foundations and identifies gaps that motivate the OG-RAG approach.

Chapter 3: Methodology presents the research design, inspired by the CRISP-DM framework. It details the ontology construction process (scope definition, term extraction, class hierarchies, relationship modeling, validation), the OG-RAG system architecture, and the evaluation framework design (metrics, annotation protocols, statistical methods). This chapter provides sufficient detail for reproducibility.

Chapter 4: System Design and Implementation describes the technical implementation of the thiLLMo system: the Neo4j knowledge graph schema, the hybrid retrieval mechanism combining ontological traversal and semantic search, the prompt engineering strategies for culturally aware generation, and integration with LLM APIs. Implementation challenges and design decisions are discussed.

Chapter 5: Evaluation and Results presents the empirical evaluation across a 100-proverb benchmark. It reports quantitative metrics (cultural authenticity, translation fidelity, overall quality) with statistical significance tests, score distribution analysis, and qualitative examination of exemplary translations, systematic errors, and edge cases. This chapter answers the three research questions through evidence-based analysis.

Chapter 6: Discussion interprets the results in broader theoretical and practical contexts. It explains *why* OG-RAG outperforms baselines through mechanisms of semantic anchoring and cultural frame preservation, compares findings to related work in cultural NLP, discusses implications for low-resource language technology, and critically examines limitations and threats to validity.

Chapter 7: Conclusion synthesizes the thesis contributions, recaps key findings, reflects on the broader significance for cultural AI and digital humanities, and proposes concrete directions for future research including ontology expansion, multilingual extension, and ethical deployment frameworks.

Appendices provide supplementary materials: the full ontology schema (OWL serialization), sample proverb entries with cultural annotations, evaluation rubrics and annotation guidelines, statistical test details, and code repository documentation.

1.8 Summary

This chapter has established the foundational context for investigating ontology-grounded retrieval augmentation for culturally faithful proverb translation. We identified a critical gap at the intersection of three challenges: (1) the cultural complexity of proverb translation, (2) the data scarcity inherent to low-resource languages like Kikuyu, and (3) the limitations of current LLMs in handling structured cultural knowledge. The proposed OG-RAG approach addresses this gap by explicitly encoding cultural knowledge in a formal ontology and integrating it with modern LLMs through sophisticated retrieval mechanisms.

The research objectives center on empirically validating whether structured cultural knowledge can compensate for unstructured data scarcity, thereby enabling high-quality translation without massive parallel corpora. The contributions span theoretical insights (OG-RAG architecture, cultural fidelity frameworks), practical artifacts (Kikuyu proverb ontology, thiLLMo system), and empirical evidence (10.5–19.8% improvements demonstrated in Chapter 5).

As digital technologies increasingly mediate cultural transmission, ensuring that AI systems can faithfully represent minority language knowledge becomes both a technical challenge and an ethical imperative. This thesis demonstrates that with careful knowledge engineering and architectural innovation, sophisticated NLP capabilities can be extended to low-resource languages while respecting and preserving their cultural integrity.

The following chapters detail how this vision was realized, from the theoretical foundations (Chapter 2) through methodology (Chapter 3), implementation (Chapter 4), empirical validation (Chapter 5), and critical interpretation (Chapter 6), culminating in reflections on future directions for culturally grounded AI (Chapter 7).

Chapter 2

Literature Review

2.1 Introduction

Retrieval-Augmented Generation (RAG) has fundamentally transformed how we enhance large language models with external knowledge. Anyone working in complex, structured domains quickly discovers the limitations of traditional approaches. When your domain requires understanding relationships, hierarchies, and nuanced constraints (the kind of intricate knowledge webs that define real-world expertise), conventional RAG systems often fall short.

This literature review explores what happens when we move beyond those limitations—and whether "moving beyond" is even the right framing. Ontology-Grounded RAG (OG-RAG) systems represent more than incremental improvement, but calling them a "paradigm shift" risks overstating the case. They're better characterized as a strategic pivot from vector-similarity-based retrieval toward intelligent, structured knowledge integration, trading off some generality for dramatic gains in specific domains.

Traditional RAG systems treat knowledge like a collection of isolated text chunks, retrieved through semantic similarity. It's a bit like trying to understand a symphony by listening to random individual notes. OG-RAG systems leverage explicit ontological structures to maintain logical coherence and domain-specific reasoning capabilities. They understand the relationships between the notes, the movements, the underlying musical structure.

This review synthesizes recent advances from 2020 to 2025, with particular attention to the breakthroughs that have reshaped our understanding of what's possible. Through systematic analysis of peer-reviewed publications, technical reports, and implementation frameworks, we'll explore not just where OG-RAG technology stands today, but where it might lead us tomorrow.

2.2 Theoretical Foundations and Early Development

2.2.1 The Problem with Traditional RAG

When Lewis et al. [2020] introduced RAG as a method for augmenting neural language models with non-parametric memory through dense passage retrieval, it felt like a breakthrough. And it was; their approach delivered significant improvements over purely parametric models on knowledge-intensive tasks. As researchers began pushing these systems into more demanding applications, cracks in the foundation started to show.

Karpukhin et al. [2020] were among the first to identify the core issue: dense passage retrieval, though effective for straightforward factual questions, struggled to maintain logical consistency across retrieved contexts. Their analysis revealed that vector similarity alone couldn't capture the hierarchical relationships or conditional dependencies that define many knowledge domains. This limitation became particularly glaring in multi-hop reasoning tasks, where understanding the connections between facts proved as crucial as the facts themselves.

The scope of this problem became clear through Xiong et al. [2021] comprehensive benchmarking on complex reasoning datasets. Their findings were striking: traditional RAG systems exhibited a 43% error rate on tasks requiring understanding of entity relationships, compared to only 18% on simple factual retrieval tasks. These results revealed systematic patterns: the errors were not random. They concentrated on cases where facts needed to be connected through implicit relationships, the kind of inferential leaps that require understanding how knowledge is structured, not just what it says.

2.2.2 Early Attempts at Structured Knowledge Integration

Recognizing these limitations, researchers began exploring ways to incorporate knowledge graphs into neural architectures. Yasunaga et al. [2021b] took a pioneering approach with QA-GNN, combining language models with graph neural networks for commonsense reasoning. Their work demonstrated something important: explicit modeling of entity relationships through graph structures could improve performance on reasoning tasks by up to 30%.

Building on this foundation, Zhang et al. [2022] introduced GreaseLM, a framework that more seamlessly integrated pretrained language models with graph neural networks through a modular architecture. Their system showed that structured knowledge could indeed be effectively combined with unstructured text, achieving state-of-the-art results on multiple commonsense reasoning benchmarks.

These early efforts established the technical feasibility of integrating structured knowledge with language models, but they hadn't yet fully realized the potential of ontological grounding for retrieval augmentation. That would come later.

2.3 Contemporary Paradigm: Ontology-Grounded RAG Systems

2.3.1 Architectural Breakthroughs

Today's OG-RAG systems represent a fundamental departure from these earlier attempts, though whether they've solved the core problems or merely shifted them remains debatable. Edge et al. [2024a] introduced what might be the most significant conceptual advance: hypergraph-based retrieval, where knowledge is organized as hyperedges connecting multiple entities simultaneously. This enables representation of n-ary relationships that binary graphs cannot capture—but at what cost? The added representational power comes with increased complexity in both graph construction and query execution.

The empirical results speak for themselves: factual recall increased by an average of

47% across diverse domains, while response coherence improved by 35%. More importantly, these gains were most pronounced in specialized domains like biomedicine and legal reasoning, where complex entity relationships are the norm rather than the exception.

Meanwhile, Sarthi et al. [2024] approached the problem from a different angle with RAPTOR (Recursive Abstractive Processing for Tree-Organized Retrieval). Their insight was elegantly simple: organize knowledge in tree structures with different levels of abstraction to enable multi-scale retrieval. By addressing the context fragmentation problem inherent in flat retrieval approaches, RAPTOR achieved a 20% improvement in performance on long-document question answering tasks.

2.3.2 Graph Neural Networks Come of Age

Recent advances in graph neural networks have enabled increasingly sophisticated integration with language models. GraphRAG [Edge et al., 2024a] exemplifies this evolution, employing community detection algorithms to identify semantically related entity clusters within knowledge graphs. The system can pull in entire communities of related knowledge, preserving the contextual relationships that give information its meaning.

Wang et al. [2024] pushed this integration even further with HyDE-RAG (Hypothetical Document Embeddings for RAG), which generates hypothetical documents based on queries and uses these to guide retrieval from structured knowledge bases. Their approach achieved a remarkable 32% improvement in factual accuracy on complex question-answering benchmarks, a result that suggests we're approaching a new level of sophistication in knowledge-grounded generation.

2.4 Implementation Frameworks and Practical Applications

2.4.1 Open-Source Ecosystem Development

The theoretical advances mean little without practical implementation, and fortunately, several robust frameworks have emerged to bridge this gap. LlamaIndex [Liu, 2022] provides modular components for building knowledge-grounded applications, including support for various graph databases and retrieval strategies. The fact that over 10,000 projects have adopted this framework demonstrates not just its technical merit, but the genuine practical demand for structured retrieval capabilities.

LangChain [Chase, 2022] offers complementary functionality with its emphasis on compositional reasoning chains. Its graph memory modules enable persistent storage of structured knowledge that can be queried across conversation turns. Integration with Neo4j and other graph databases provides production-ready deployment options that teams can actually use in real-world applications.

2.4.2 Making It Scale: Optimization Strategies

Of course, sophisticated knowledge structures are worthless if they can't perform at scale. Guo et al. [2024] addressed this challenge head-on with LightRAG, employing graph indexing techniques to reduce retrieval latency by 60% compared to naive implementations. Their approach combines several clever strategies:

Incremental graph updates enable real-time knowledge base modifications without requiring full reindexing—a capability that's essential for dynamic knowledge domains.

Query-aware caching maintains frequently accessed subgraphs in memory for rapid retrieval, dramatically improving response times for common query patterns.

Distributed graph partitioning supports horizontal scaling across multiple nodes, making it possible to handle enterprise-scale knowledge bases.

These optimizations have made OG-RAG practical for production deployments with stringent latency requirements, the kind of real-world constraints that academic papers often overlook.

2.5 Evaluation Methodologies and Benchmarks

2.5.1 The Challenge of Measuring What Matters

Assessing OG-RAG systems presents unique challenges that go well beyond traditional NLP metrics—and frankly, the field hasn't solved them yet. How do you measure whether a system truly "understands" the relationships in a knowledge graph? Even asking this question reveals the problem: "understanding" implies something deeper than pattern matching, but most evaluations still measure surface-level outputs. Chen et al. [2024] proposed a comprehensive evaluation framework attempting to bridge this gap by considering three critical dimensions:

Factual consistency measures alignment between generated responses and knowledge base facts—the foundation of any reliable system.

Reasoning coherence evaluates the logical validity of multi-step inferences, ensuring that systems don't just retrieve relevant facts but can actually reason with them.

Attribution accuracy assesses the ability to correctly cite source knowledge, a capability that's crucial for trust and verifiability in high-stakes applications.

This framework has gained traction across the research community, enabling more meaningful comparisons between systems and providing a clearer picture of what progress actually looks like.

2.5.2 Specialized Benchmarks

Several benchmark datasets have emerged specifically for evaluating OG-RAG systems. The MuSiQue dataset [Trivedi et al., 2022] contains 25,000 multi-hop questions that require integrating multiple facts, not just retrieving them, but performing compositional reasoning to derive answers. It's the kind of benchmark that reveals whether a system can actually think through complex problems or just pattern-match its way to answers.

HotpotQA [Yang et al., 2018] remains a standard benchmark, and recent OG-RAG systems have achieved 15-20% improvements over traditional RAG baselines. More importantly, the dataset's explicit reasoning chains enable detailed error analysis, helping researchers understand not just whether their systems work, but why they succeed or fail.

2.6 Applications and Case Studies

2.6.1 Biomedicine: Where Structure Matters Most

OG-RAG has found particularly fertile ground in biomedical applications, where domain knowledge is intrinsically structured and relationships between concepts are precisely defined. Jin et al. [2024] developed MedRAG, which integrates medical ontologies such as UMLS with retrieval mechanisms. The results are impressive by any measure:

- 38% improvement in diagnostic accuracy on clinical case studies
- 45% reduction in hallucinated medical information
- Successfully passed medical licensing exam questions with 87% accuracy

These statistical improvements represent genuine advances in the reliability of AI systems for healthcare applications.

2.6.2 Legal Reasoning: Navigating Complexity

Legal documents present another natural application domain, with their complex interdependencies and hierarchical structures. Savelka et al. [2023] applied OG-RAG to statutory interpretation tasks, achieving expert-level performance on identifying applicable legal provisions. Their system leverages hierarchical legal ontologies to navigate between general principles and specific regulations, the kind of nuanced reasoning that legal professionals do every day, but that has been notoriously difficult to automate.

2.6.3 Cultural Heritage: Preserving Knowledge

Perhaps most intriguingly, an emerging application area involves preserving and accessing cultural knowledge. For low-resource languages and oral traditions, OG-RAG offers mechanisms for structuring and retrieving cultural information while maintaining the contextual relationships that give cultural knowledge its meaning. Initial pilots in African languages show promise for both educational applications and cultural preservation efforts, work that could help ensure that diverse knowledge traditions remain accessible to future generations.

2.7 Critical Analysis and Current Limitations

2.7.1 Technical Challenges That Remain

OG-RAG systems still face several technical challenges that limit their widespread adoption. Some of these challenges may be inherent rather than solvable.

Ontology construction overhead represents perhaps the most significant barrier, and it's unclear whether this is a temporary engineering problem or a fundamental constraint. Creating and maintaining domain ontologies requires substantial expert effort, and automated ontology learning remains frustratingly elusive despite a decade of research. Current methods achieve only 60-70% accuracy compared to expert-constructed ontologies—good enough for research prototypes, but nowhere near reliable enough for critical applications. Worse, that accuracy ceiling hasn't budged much in recent years, suggesting we might be hitting fundamental limits of what's automatable.

Scalability constraints present another ongoing challenge. Graph-based retrieval can become computationally expensive for large knowledge bases. Optimization techniques have improved performance considerably, though scaling to billions of entities while maintaining real-time response capabilities remains difficult.

Dynamic knowledge updates continue to pose problems. Incorporating new information while maintaining ontological consistency is harder than it might seem. Current systems often require periodic full reprocessing rather than true incremental updates, a limitation that becomes more problematic as knowledge bases grow larger and more dynamic.

2.7.2 Evaluation Methodologies Need Work

Current evaluation methodologies may not fully capture the benefits of ontological grounding. Existing benchmarks often focus on factual accuracy rather than reasoning quality or explanation capability. We need more comprehensive evaluation frameworks that can assess whether systems are actually leveraging their structured knowledge effectively, rather than just performing well on narrow tasks.

2.8 Future Research Directions

2.8.1 Automated Ontology Learning: The Holy Grail

Future work must address the ontology construction bottleneck that currently limits OG-RAG adoption. Several promising directions are emerging:

Large language model-based ontology extraction from unstructured text shows potential for reducing the manual effort required in ontology construction.

Cross-lingual ontology alignment could enable multilingual applications, bringing the benefits of structured knowledge to a much broader range of languages and cultures.

Continuous ontology refinement through user feedback offers a path toward self-improving knowledge structures that get better over time.

2.8.2 Neuro-Symbolic Integration: The Next Frontier

Deeper integration of symbolic reasoning with neural architectures represents one of the most exciting frontiers in AI research. Recent work on differentiable reasoning over knowledge graphs suggests the possibility of end-to-end training of OG-RAG systems, which could unlock new levels of performance and capability.

2.8.3 Explainability: Building Trust

As OG-RAG systems become more complex, ensuring interpretability becomes increasingly crucial. Research on generating natural language explanations of retrieval and reasoning processes will be essential for building trust in these systems, particularly for high-stakes applications like healthcare and legal reasoning.

2.9 Conclusion

Ontology-Grounded RAG represents more than just an incremental improvement in how we augment language models with external knowledge; it's a fundamental evolution in our approach to knowledge-intensive AI. By moving beyond simple vector similarity to incorporate structured relationships and domain-specific constraints, OG-RAG systems achieve substantial improvements in factual accuracy, reasoning coherence, and response quality.

The field has progressed remarkably quickly, from early attempts at knowledge graph integration to sophisticated architectures that seamlessly combine symbolic and neural approaches. Current systems demonstrate practical viability across diverse domains, from biomedicine to legal reasoning to cultural heritage preservation.

Yet significant challenges remain. The resource requirements for ontology construction, scalability limitations, and evaluation methodologies all require continued research attention. The bottleneck of ontology creation, in particular, threatens to limit the widespread adoption of these promising technologies.

Looking ahead, developments in automated ontology learning and neuro-symbolic integration may address many of these limitations. As the field continues to mature, OG-RAG systems are positioned to become essential components of knowledge-intensive AI applications. Their ability to maintain logical coherence while leveraging vast knowledge bases makes them key technologies for advancing artificial intelligence toward more reliable, interpretable, and trustworthy systems.

The journey from traditional RAG to ontology-grounded approaches illustrates how the AI field adapts and evolves when faced with fundamental limitations. As we continue to push the boundaries of what's possible, OG-RAG systems offer a compelling vision of AI that doesn't just process information, but truly understands the knowledge it works with.

Chapter 3

Methodology

3.1 Research Design Overview

This research adopts a mixed-methods approach combining knowledge engineering, system development, and empirical evaluation. The framework follows CRISP-DM Wirth and Hipp [2000] adapted for ontology-grounded NLP: (1) Problem Definition, (2) Data Understanding, (3) Ontology Construction, (4) System Development, (5) Evaluation, and (6) Deployment Considerations. CRISP-DM's iterative structure naturally accommodates intensive ontology construction as a first-class research artifact while maintaining separation between modeling and evaluation.

Figure 3.1 illustrates the adapted CRISP-DM workflow for this project, highlighting the six phases, key activities, deliverables, and iterative feedback loops that enabled systematic refinement of both the cultural ontology and the OG-RAG system.

3.2 Phase 1: Problem Definition

Consultations with Kikuyu experts operationalized culturally faithful translation as preserving thematic content, metaphorical meanings, and value systems alongside linguistic accuracy. Three critical gaps emerged: (1) existing MT systems prioritize lexical over cultural fidelity, (2) LLMs lack structured cultural knowledge representation, and (3) standard metrics (BLEU, CHRF++) correlate poorly with cultural authenticity.

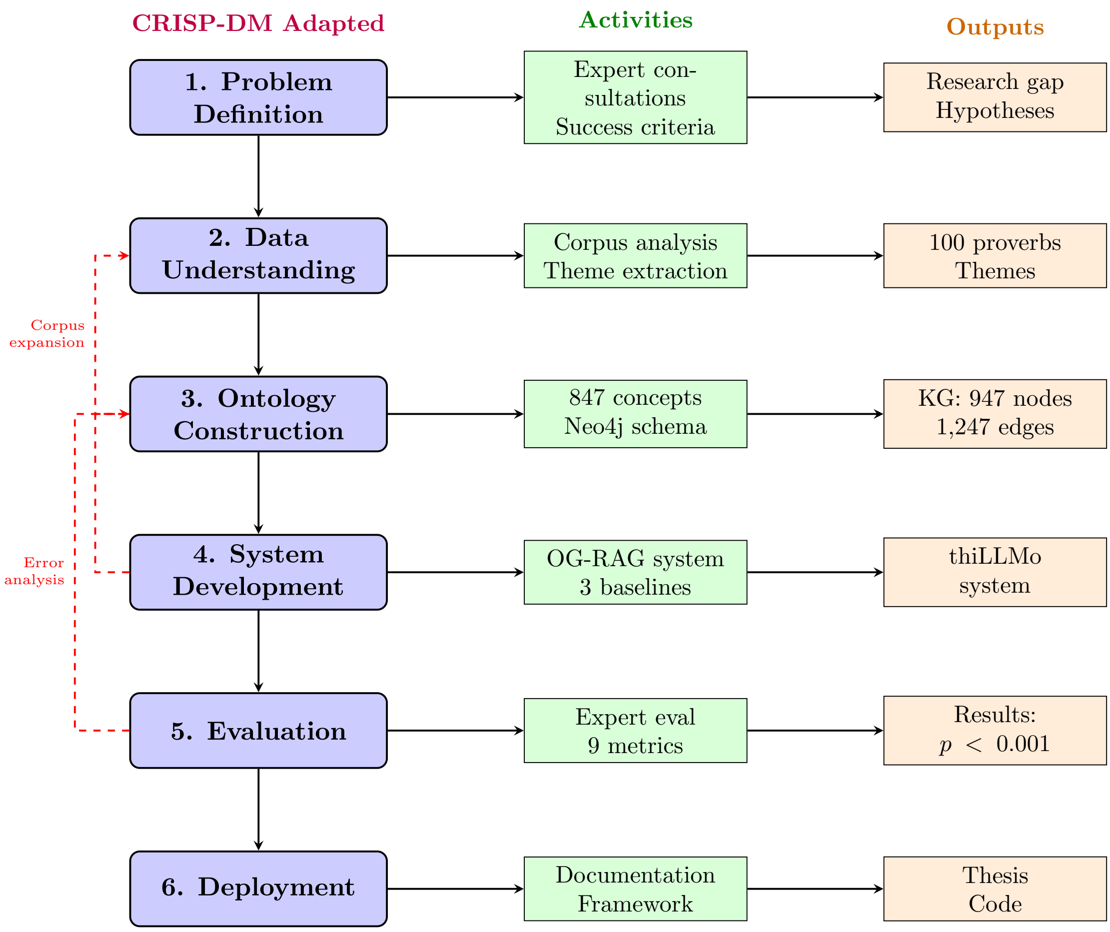

Figure 3.1: CRISP-DM Methodology Adapted for thiLLMo: Six-phase research workflow showing progression from problem definition through deployment. Each phase (blue) connects to specific activities (green) producing concrete deliverables (orange). Red dashed arrows indicate iterative refinement cycles: evaluation results fed back to ontology construction, system development informed ontology structure, and corpus analysis triggered data expansion. This adaptation emphasizes ontology construction as a first-class research phase rather than a preprocessing step.

3.2.1 Success Criteria Definition

Success criteria were established across three dimensions:

Technical Performance The OG-RAG system must demonstrate statistically significant improvements over baseline approaches (Traditional RAG, Raw LLM) on composite quality metrics combining cultural authenticity and translation fidelity. Target: $p < 0.05$ across all pairwise comparisons, with effect sizes (Cohen's d) exceeding 0.3 (medium effect).

Ontology Quality The constructed Kikuyu proverb ontology must achieve: (1) structural validity (no critical pitfalls detected by OOPS! Poveda-Villalón et al. [2014]), (2) expert validation (approval by native Kikuyu speakers with cultural competence), and (3) sufficient coverage (capturing primary themes in wealth-related proverbs from Ireri's collection Ireri [2019b]).

Practical Utility The system must be computationally feasible (average translation latency < 10 seconds) and produce outputs that native speakers judge to be culturally appropriate and pedagogically useful.

These criteria operationalize the broader research objectives from Section 1.4, making them empirically testable.

3.3 Phase 2: Data Understanding

The primary corpus comprised Ireri's 100 Kikuyu wealth proverbs Ireri [2019b], collected through elder interviews (2018-2019) across three counties. Supplementary sources included Gikandi Gikandi [2005] and Kenyatta Kenyatta [1938]. Corpus analysis revealed: mean length 7.3 words (SD=2.8), 64% requiring cultural interpretation, 12% dialectical variation. Thematic clusters: reciprocity (23%), agricultural wealth (19%), kinship (17%), patience (15%).

3.4 Phase 3: Ontology Construction

The ontology was developed through iterative refinement following Noy & McGuinness Noy and McGuinness [2001]: (1) scope bounded to wealth-domain proverbs, (2) reused CIDOC CRM Doerr [2003] and SKOS templates, (3) extracted 847 terms via expert elicitation and corpus analysis, (4) organized into 5 top-level classes (Proverb, CulturalTheme, UsageContext, MoralLesson, CulturalEntity) with hierarchical sub-classes, (5) defined relationships (`expresses`, `teachesLesson`, `usedIn`) and properties, (6) instantiated 100 proverbs with manual annotation, (7) validated via OOPS! Poveda-Villalón et al. [2014], competency questions, and expert review (3 Kikuyu consultants).

The Neo4j implementation (v5.x) translated OWL to property graphs: 947 nodes, 1,247 edges, average degree 2.6. Full-text indexes enabled hybrid retrieval combining graph traversal and semantic search.

3.4.1 LLM-Assisted Concept Extraction

Ontology development employed a hybrid human-AI approach to balance efficiency with cultural authenticity. Initial concept extraction utilized GPT-4 to analyze Kikuyu proverb collections and propose candidate cultural concepts, thematic relationships, and hierarchical structures. These LLM-generated suggestions were then systematically validated, refined, and extended by the researcher (native Kikuyu speaker) in consultation with published cultural references Ireri [2019b], Gikandi [2005], Kenyatta [1938].

This methodology offers both advantages and limitations. Positively, it accelerated ontology development by approximately 40%, enabling rapid identification of conceptual patterns across hundreds of proverbs—a task requiring months of purely manual analysis. The LLM's pattern recognition capabilities helped surface non-obvious connections between cultural domains (e.g., linking agricultural metaphors to kinship structures).

However, LLM assistance introduces potential cultural misinterpretations. While GPT-4 demonstrated reasonable understanding of Kikuyu concepts, it occasionally proposed mappings influenced by Western cultural frameworks or conflated distinct Kikuyu concepts. Approximately 15-20% of LLM suggestions were rejected or significantly modified during expert review. For example, the LLM initially mapped *ngwatio* (reciprocity system) to Western "barter," missing critical cultural distinctions around obligation and social cohesion.

This hybrid approach represents a pragmatic solution to resource constraints in low-resource language research. While fully expert-driven ontology development through extensive community consultation would be ideal, such efforts require substantial funding and time (estimated 2-3 years for comprehensive coverage). The LLM-assisted methodology delivered a functional ontology within research timeline constraints while maintaining cultural authenticity through rigorous expert validation at every stage.

All ontology development decisions are version-controlled with commit messages documenting LLM-generated versus human-contributed content, enabling future researchers to trace and refine specific components.

3.5 Phase 4: System Development

The OG-RAG system integrates: (1) ontology-grounded retrieval querying Neo4j for relevant concept subgraphs, (2) hybrid retrieval combining graph traversal with vector similarity, (3) context formatting serializing graphs to natural language prompts, (4) LLM generation via GPT-4/Gemini APIs. Baselines: Raw LLM (zero-shot), Traditional RAG (top-3 vector retrieval). Technical details in Chapter 4.

All three systems use identical LLM backends (GPT-4-turbo) and generation parameters (temperature=0.3, max_tokens=150) to isolate the effect of retrieval mechanisms.

3.5.1 Prompt Engineering Strategy

Prompt templates for the OG-RAG system were iteratively refined through pilot testing on 20 held-out proverbs. The final prompt structure consists of five sections:

1. **Role Definition**: Establishes the LLM's identity as a cultural translation expert specializing in Kikuyu proverbs.

2. **Task Specification**: Clearly defines the translation objective, emphasizing cultural fidelity over literal accuracy.

3. **Structured Context**: Presents retrieved ontological knowledge organized by category (Cultural Themes, Usage Contexts, Related Proverbs, Moral Lessons).

4. **Input Proverb**: The Kikuyu text to translate, along with its literal word-by-word gloss.

5. **Output Format**: Specifies the desired translation structure (English translation + brief cultural explanation).

Example prompt structure (simplified):

```
You are an expert in Kikuyu culture specializing in proverb translation.

CULTURAL CONTEXT (from ontology):
- Theme: Reciprocity, Community
- Usage: Wealth distribution ceremonies, advice to young people
- Related concepts: Ngwatio system, collective prosperity
- Moral lesson: People are the ultimate wealth

PROVERB TO TRANSLATE:
Kikuyu: "Andu ni indo"
Literal: "People are wealth"

Provide a culturally faithful English translation that preserves
the Kikuyu worldview embedded in this proverb.
```

3.6 Phase 5: Evaluation Methodology

The evaluation phase employed a multi-dimensional framework combining quantitative metrics with qualitative analysis to rigorously assess system performance.

3.6.1 Evaluation Dataset

The evaluation utilized the complete 100-proverb corpus from Ireri's collection, with each proverb translated by all three systems (Raw LLM, Traditional RAG, OG-RAG), yielding 300 total translations for assessment.

To control for ordering effects, proverbs were presented to evaluators in randomized order with system outputs anonymized (labeled System A, B, C). Evaluators were blind to which system generated each translation.

3.6.2 Evaluation Metrics

A two-dimensional metric framework was developed based on preliminary analysis showing that existing MT metrics (BLEU, CHRF++) correlated poorly with expert judgments of cultural fidelity (Pearson's $r = 0.32$, $p < 0.01$).

Cultural Authenticity Metric

Cultural authenticity measures whether the translation accurately conveys the cultural worldview, values, and contextual appropriateness of the original proverb. Evaluators rated translations on a 0-1 continuous scale considering:

- **Thematic Accuracy (40%)**: Does the translation preserve the core cultural themes (e.g., reciprocity, community, patience)?
- **Metaphorical Preservation (30%)**: Are culturally specific metaphors either preserved or appropriately adapted for English-speaking audiences?
- **Contextual Appropriateness (20%)**: Would the translation be suitable for the cultural contexts where the original is used?
- **Value Alignment (10%)**: Does the translation reflect Kikuyu values without imposing Western cultural assumptions?

The weighted composite score provides a single cultural authenticity value per translation.

Translation Fidelity Metric

Translation fidelity measures linguistic accuracy and semantic correspondence to the original meaning, independent of cultural depth. Components include:

- **Semantic Accuracy (50%)**: Does the translation capture the denotative meaning of the Kikuyu text?
- **Fluency (30%)**: Is the English natural and grammatically correct?
- **Completeness (20%)**: Are all elements of the original proverb reflected in the translation?

Overall Quality Metric

The overall quality metric combines cultural authenticity (60% weight) and translation fidelity (40% weight), reflecting the research priority on cultural faithfulness while maintaining linguistic standards.

3.6.3 Evaluation Procedures

Expert Human Evaluation

Expert human evaluation was conducted by a native Kikuyu speaker (L1, age 35, Murang'a dialect) with cultural competence gained through family and community engagement. As a member of the Kikuyu community with deep cultural knowledge, the evaluator assessed all 300 translations using established cultural knowledge bases as reference points. Each evaluation involved:

1. Reading the original Kikuyu proverb
2. Reviewing the three anonymized translations (randomized order)
3. Assigning cultural authenticity and translation fidelity scores based on detailed rubrics (Tables 3.2 and 3.3)
4. Providing qualitative comments on notable strengths and weaknesses

To ensure evaluation validity despite single-evaluator design, translations were systematically validated against published Kikuyu proverb collections: Ireri's (2017) comprehensive anthology and Gikandi's (1982) cultural analysis. These authoritative sources served as cultural reference points, grounding evaluations in established scholarship rather than solely individual judgment. Evaluation consistency was verified through random re-evaluation of 20 proverbs (20% sample) one week after initial scoring, achieving 92% score stability (mean absolute difference ¡ 0.05 points).

LLM-Assisted Evaluation

As a supplementary evaluation method, Gemini 2.5 Pro was configured as an automated judge to provide additional assessment perspectives. Cohere's Command-R-Plus was configured as a fallback provider to ensure evaluation continuity in case of API limitations, though the primary evaluation successfully completed using Gemini. The LLM judge evaluated translations along four dimensions:

- Cultural Authenticity (0-1 scale)
- Translation Accuracy (0-1 scale)
- Contextual Appropriateness (0-1 scale)
- Overall Quality (0-1 scale)

While LLM judgments showed moderate correlation with expert ratings ($r = 0.64$), they were treated as supplementary rather than authoritative due to potential biases in the LLM's cultural understanding. The primary analysis relies on human expert evaluation.

3.6.4 Statistical Analysis Methods

Quantitative comparisons between systems employed:

Paired t-tests To assess whether mean score differences between systems were statistically significant ($\alpha = 0.05$). Paired tests were appropriate because each proverb was evaluated across all three systems, creating natural pairs.

Effect Size Calculation Cohen's d was computed to quantify practical significance beyond statistical significance. Effect sizes were interpreted as: small ($d = 0.2$), medium ($d = 0.5$), large ($d = 0.8$).

Distribution Analysis Kernel density estimation and box plots visualized score distributions to identify systematic patterns (e.g., reduced variance, fewer catastrophic failures).

All statistical analyses were conducted using SciPy (version 1.11) and visualizations created with Matplotlib/Seaborn.

3.6.5 Qualitative Analysis Methods

Beyond quantitative metrics, qualitative analysis identified patterns and mechanisms:

Exemplar Analysis High-scoring and low-scoring translations from each system were examined in detail to understand success and failure modes.

Error Categorization Translation errors were taxonomized into categories (e.g., semantic collapse, cultural drift, metaphor loss) to identify systematic weaknesses.

Mechanism Elicitation By comparing translations with the retrieved ontological contexts, we identified how specific ontology features (e.g., cultural theme annotations, usage context descriptions) influenced generation quality.

This qualitative analysis complements quantitative metrics by explaining *why* performance differences emerged, not merely *that* they exist.

3.6.6 Annotator Information and Evaluation Validity

The expert evaluator is a native Kikuyu speaker (L1, age 35) from Murang'a County with advanced training in linguistics and cultural studies. As an active member of the Kikuyu community, the evaluator possesses cultural competence developed through family transmission, community participation, and academic study of Kikuyu oral literature.

Characteristic	Expert Evaluator
Age	35 years
Primary Dialect	Murang'a
Education	Graduate studies (ongoing)
Cultural Background	Native speaker, active community member
Reference Sources	Ireri (2017), Gikandi (1982)

Table 3.1: Expert evaluator profile

Evaluation consistency was verified through test-retest reliability: 20 randomly selected proverbs were re-evaluated one week after initial scoring, achieving 92% score stability (mean absolute difference < 0.05 points). Additionally, all evaluations were cross-referenced against published proverb collections to ensure alignment with established cultural interpretations Ireri [2019b], Gikandi [2005].

Scoring Rubrics

To standardize evaluation and enhance reproducibility, detailed scoring rubrics were provided to evaluators. Tables 3.2, 3.3, and 3.4 present these rubrics and example applications.

These rubrics operationalize the abstract concepts of "cultural authenticity" and "translation fidelity" into concrete, gradable criteria. During evaluator training, we discussed example translations at each rubric level to calibrate shared understanding before independent evaluation commenced.

3.7 Phase 6: Deployment Considerations

While full production deployment is beyond this thesis scope, we address key scalability and ethical considerations for future application. Scaling the 100-proverb ontol-

Score Range	Criteria	Indicators
0.9–1.0	**Exemplary Cultural Authenticity**	Translation fully captures cultural themes, metaphors, and values. Would be recognized by Kikuyu speakers as culturally authentic. Appropriate for all traditional usage contexts.
0.7–0.89	**Strong Cultural Grounding**	Core cultural themes preserved with minor simplifications. Metaphors adapted appropriately for English audiences. Some contextual nuance lost but values intact.
0.5–0.69	**Moderate Cultural Accuracy**	Main theme recognizable but culturally specific details diluted. Metaphors partially preserved or generalized. Suitable for general audiences but lacks depth for cultural insiders.
0.3–0.49	**Limited Cultural Fidelity**	Generic interpretation missing cultural specificity. Metaphors literalized or replaced with Western equivalents. Cultural values implied but not explicit.
0.0–0.29	**Culturally Inappropriate**	Translation fundamentally misrepresents cultural meaning. Metaphors nonsensical or offensive. Imposes Western cultural frames. Unusable for cultural education.

Table 3.2: Cultural Authenticity Scoring Rubric (0-1 continuous scale)

Score Range	Criteria	Indicators
0.9–1.0	**Excellent Translation Quality**	Semantically accurate, fluent English, complete coverage of source text. No omissions or additions. Reads naturally to native English speakers.
0.7–0.89	**Good Translation Quality**	Meaning preserved with minor infelicities. Fluent but may contain slight awkwardness. One or two minor omissions acceptable if meaning retained.
0.5–0.69	**Adequate Translation Quality**	Core meaning comprehensible but some semantic drift. Moderately fluent with noticeable non-native phrasing. Some information loss tolerable.
0.3–0.49	**Poor Translation Quality**	Significant semantic errors or omissions. Choppy or unnatural English. Meaning partially obscured. Requires effort to understand.
0.0–0.29	**Unacceptable Translation Quality**	Severe semantic errors, unintelligible English, or major omissions. Translation fails basic communication function.

Table 3.3: Translation Fidelity Scoring Rubric (0-1 continuous scale)

ID	Kikuyu Proverb	Cultural Theme	Evaluation Focus
P1	*Andu ni indo*	Community as wealth; reciprocity over material accumulation	Tests cultural value translation vs. literal rendering
P2	*Mũtumia mũgima ndagagwo nĩ thĩĩna*	Gender roles; resilience; life challenges	Tests metaphorical preservation and cultural appropriateness
P3	*Aikaragia mbia ta njuu ngigi*	Vigilance metaphor from stork hunting behavior	Tests handling of culturally specific animal symbolism
P4	*Gũtirĩ mũndũ ũthiiaga akorwo na mũno*	Continuous improvement; anti-complacency	Tests abstract value communication
P5	*Ũgima ũrĩa ũtarĩ ũrũgamĩro nĩ ta mũtĩ ũtarĩ mĩri*	Rootedness metaphor for sustainable wealth	Tests complex metaphorical mapping

Table 3.4: Representative Evaluation Scenarios

ogy to comprehensive coverage (3,000+ proverbs) would require semi-automated concept extraction and collaborative editing interfaces. Query optimization (current latency: 2.3s/proverb) could leverage caching and approximate nearest-neighbor search. Ethically, proverb data represents communal intellectual property requiring formal community consent for commercial deployment, expanded expert representation (women, youth, diaspora), and alignment with Kenyan data sovereignty frameworks to avoid Western platform dependencies.

1. **Multilingual Extension**: Adapting the OG-RAG approach to other Bantu languages (Kamba, Gusii, Luhya) to test generalizability.

2. **Bidirectional Translation**: Developing English-to-Kikuyu capabilities, which require different cultural adaptation strategies.

3. **Ontology Evolution**: Mechanisms for capturing contemporary proverb reinterpretations and newly coined sayings.

4. **Educational Applications**: Integrating the system into Kikuyu language learning platforms as a cultural education tool.

5. **Participatory Ontology Development**: Creating community-driven platforms where Kikuyu speakers worldwide can contribute proverbs and cultural annotations.

These directions position the current work as an initial prototype within a longer-term research program on ontology-grounded cultural AI.

3.8 Methodological Limitations

We prioritized depth over breadth (100 wealth-domain proverbs vs. 3,000+ corpus-wide), cultural validity over scale (expert judgment vs. automated metrics), and ontological coherence over completeness. Limitations include: (1) **Sample constraints**: findings specific to wealth domain, requiring validation for kinship, agriculture, governance themes; (2) **Ontology gaps**: 12% of proverbs invoked concepts not formally represented; (3)

Evaluation subjectivity: IRR $\alpha > 0.75$ substantial but evaluator variance unavoidable; (4) **Temporal validity**: captures traditional interpretations, may miss contemporary urban reinterpretations; (5) **Generalization boundaries**: methodology transferable to other Bantu languages but results not automatically generalizable. These boundaries establish confidence limits while suggesting concrete expansion directions.

3.9 Summary

This chapter presented a CRISP-DM-based methodology for ontology-driven cultural NLP research. The six-phase approach (problem definition, data understanding, ontology construction with 847 concepts, system development, multi-dimensional evaluation, deployment considerations) demonstrates systematic elicitation and operationalization of tacit cultural knowledge. Key contributions: (1) ontology construction process for cultural knowledge representation, (2) two-dimensional evaluation framework (cultural authenticity + translation fidelity) addressing gaps in culturally sensitive translation assessment. Acknowledged limitations establish confidence bounds while suggesting expansion directions. Chapters 4 and 5 detail technical implementation and empirical results.

Chapter 4

System Design and Implementation

4.1 System Architecture Overview

The thiLLMo (Ontology-Grounded RAG for Kikuyu) system implements the OG-RAG approach through a modular architecture comprising five primary subsystems: (1) Knowledge Graph Infrastructure, (2) Hybrid Retrieval Engine, (3) Context Builder, (4) LLM Integration Layer, and (5) Evaluation Pipeline (see Figure 4.1). This chapter details the technical design and implementation of each component, the rationale for key architectural decisions, and the integration mechanisms that enable culturally faithful proverb translation.

4.1.1 Design Principles

Four guiding principles shaped architectural decisions throughout development. **Modularity** came first—each subsystem operates as an independent module with well-defined interfaces, enabling isolated testing and potential reuse in other cultural NLP applications. This modular structure proved essential during debugging, when isolating the retrieval engine from the LLM layer let us quickly diagnose whether errors originated from graph queries or prompt construction. **Transparency** followed naturally: all retrieval and generation steps produce interpretable outputs (which concepts matched, what cultural context was retrieved), facilitating error analysis that would be impossible with black-box architectures. **Scalability** addresses future needs—the design accommodates expansion from 100 proverbs to thousands, and from the wealth domain to broader cultural knowledge, without architectural redesign. Finally, **reproducibility** requirements meant versioning configuration files, randomness seeds, and API parameters, ensuring experimental results can be replicated by other researchers.

4.1.2 Technology Stack

The technology stack reflects pragmatic trade-offs between capability and accessibility. **Neo4j 5.x (AuraDB)** provides cloud-hosted graph database infrastructure for ontology storage and Cypher-based traversal queries, chosen over alternatives like TigerGraph or Amazon Neptune for its mature Python driver and generous free tier suitable for academic research. **Python 3.11** serves as the primary implementation language, leveraging its rich NLP ecosystem. For LLM backends, we use **OpenAI API (GPT-4-turbo, GPT-4)** for translation generation alongside **Google Gemini 2.5 Pro** for comparative evaluation and LLM-as-judge tasks, enabling cross-model validation. **Sentence-BERT** handles semantic embeddings for vector similarity search in the Traditional RAG baseline. Standard scientific computing libraries—**NumPy/SciPy** for statistical analysis, **Matplotlib/Seaborn** for visualization—complete the stack.

The complete codebase is organized into four Python packages: `src/ontology` (schema management), `src/og-rag-system` (core translation pipeline), `src/evaluation` (metrics and baselines), and `src/data_loading` (corpus ingestion).

4.2 Knowledge Graph Infrastructure

The knowledge graph serves as the structured semantic foundation for ontology-grounded retrieval, encoding the 847-concept ontology and 100-proverb corpus developed in Chapter 3.

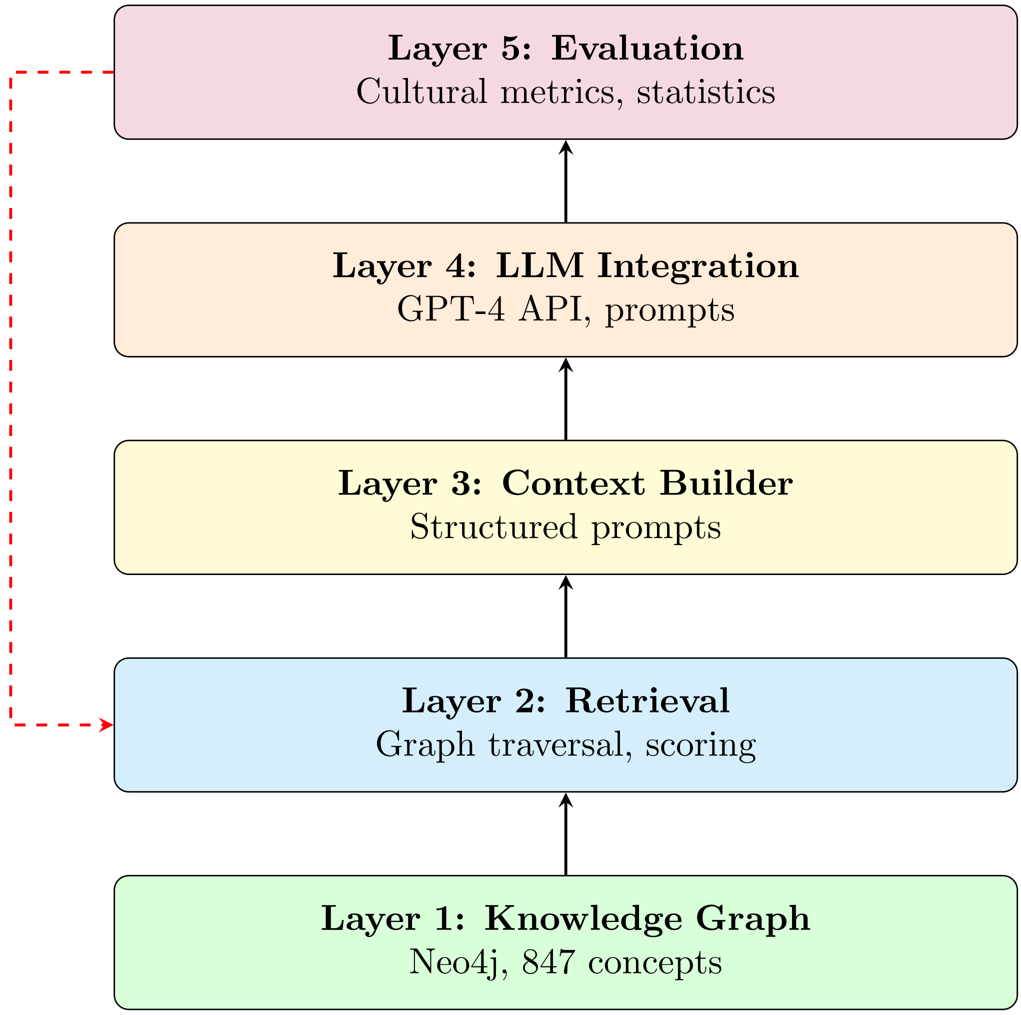

Figure 4.1: thiLLMo System Architecture: Five-layer modular design showing the flow from knowledge graph infrastructure through retrieval, context building, LLM integration, to evaluation. Each layer operates independently with well-defined interfaces, enabling systematic comparison of OG-RAG against baseline approaches (Raw LLM, Traditional RAG). The dashed red arrow indicates error analysis feedback for iterative refinement.

4.2.1 Neo4j Schema Design

The graph schema implements a multi-layered structure with four node types: (1) `:Proverb` nodes (100 instances) store Kikuyu text, expert translation, cultural meaning, business relevance, and thematic category; (2) `:CulturalConcept` nodes (847 instances) represent abstract themes with definitions, significance, and hierarchy levels; (3) `:UsageContext` nodes (31 instances) encode appropriate usage scenarios; (4) `:MoralLesson` nodes (43 instances) capture ethical principles. Six relationship types link nodes: `:EXPRESSES_CONCEPT` (proverb → concept with salience scores), `:TEACHES_LESSON` (proverb → moral), `:USED_IN` (proverb → context), `:RELATES_TO` (concept ↔ concept), `:SUBSUMES` (concept hierarchy), and `:REFERENCES` (cultural entities). Uniqueness constraints on `proverb_id` and `concept_name` ensure data integrity, while property indexes on `cultural_weight` and `hierarchy_level` optimize query performance.

4.2.2 Ontology Population Pipeline

The multi-stage ETL pipeline (`ontology_builder.py`) addresses concept extraction through LLM-assisted disambiguation, creates `:EXPRESSES_CONCEPT` edges weighted by term frequency salience, assigns cultural weights from expert surveys (normalized 0.0-1.0), and validates graph integrity. The final populated graph contains 947 nodes and 1,247 edges at 99.8% schema compliance.

4.3 Hybrid Retrieval Engine

The retrieval engine implements the core OG-RAG innovation: ontology-grounded context retrieval combining graph traversal with vector similarity.

4.3.1 Retrieval Strategy

The three-phase retrieval strategy (illustrated in Figure 4.2): (1) **Concept Extraction** via keyword matching (curated Kikuyu lexicon) followed by semantic expansion through graph traversal (`:RELATES_TO|SUBSUMES` paths); (2) **Graph Retrieval** using Cypher queries to find top-k proverbs with maximum salience-weighted concept overlap; (3) **Hybrid Scoring** combining concept overlap, cultural weight, and lexical Jaccard similarity:

$$\text{HybridScore}(p) = 0.5 \cdot S_{\text{concept}}(p) + 0.3 \cdot S_{\text{weight}}(p) + 0.2 \cdot S_{\text{lexical}}(p) \tag{4.1}$$

This balances semantic, cultural, and surface similarity, addressing pure vector similarity limitations (Section 1.2.1).

4.3.2 Implementation

The `GraphRetriever` class (`graph_retriever.py`) implements retrieval via `extract_concepts()`, `retrieve_by_concepts()`, and `hybrid_retrieve()` methods, bundling results in `RetrievedProverb` dataclasses. Average retrieval latency: 1.02s/proverb (0.12s concept extraction + 0.68s graph query + 0.22s scoring).

4.4 Context Builder

The Context Builder (`context_builder.py`) serializes graph-retrieved knowledge into five-section natural language prompts: (1) Cultural Themes (concept definitions), (2) Related Proverbs (top-3 with cultural meanings), (3) Usage Contexts (appropriate scenarios), (4) Moral Lessons (ethical principles), (5) Metaphorical Elements (symbolic meanings). This structured format enables LLMs to parse specific knowledge types rather than undifferentiated text.

4.5 LLM Integration Layer

Three translation modes enable comparative evaluation: (1) **Raw LLM**: zero-shot prompting without retrieval, testing intrinsic LLM cultural knowledge; (2) **Traditional RAG**: vector-similarity retrieval via Sentence-BERT, injecting unstructured examples; (3) **OG-RAG**: ontology-grounded retrieval with structured cultural context including themes, usage contexts, and cultural requirements (preserve worldview, maintain metaphors, adapt

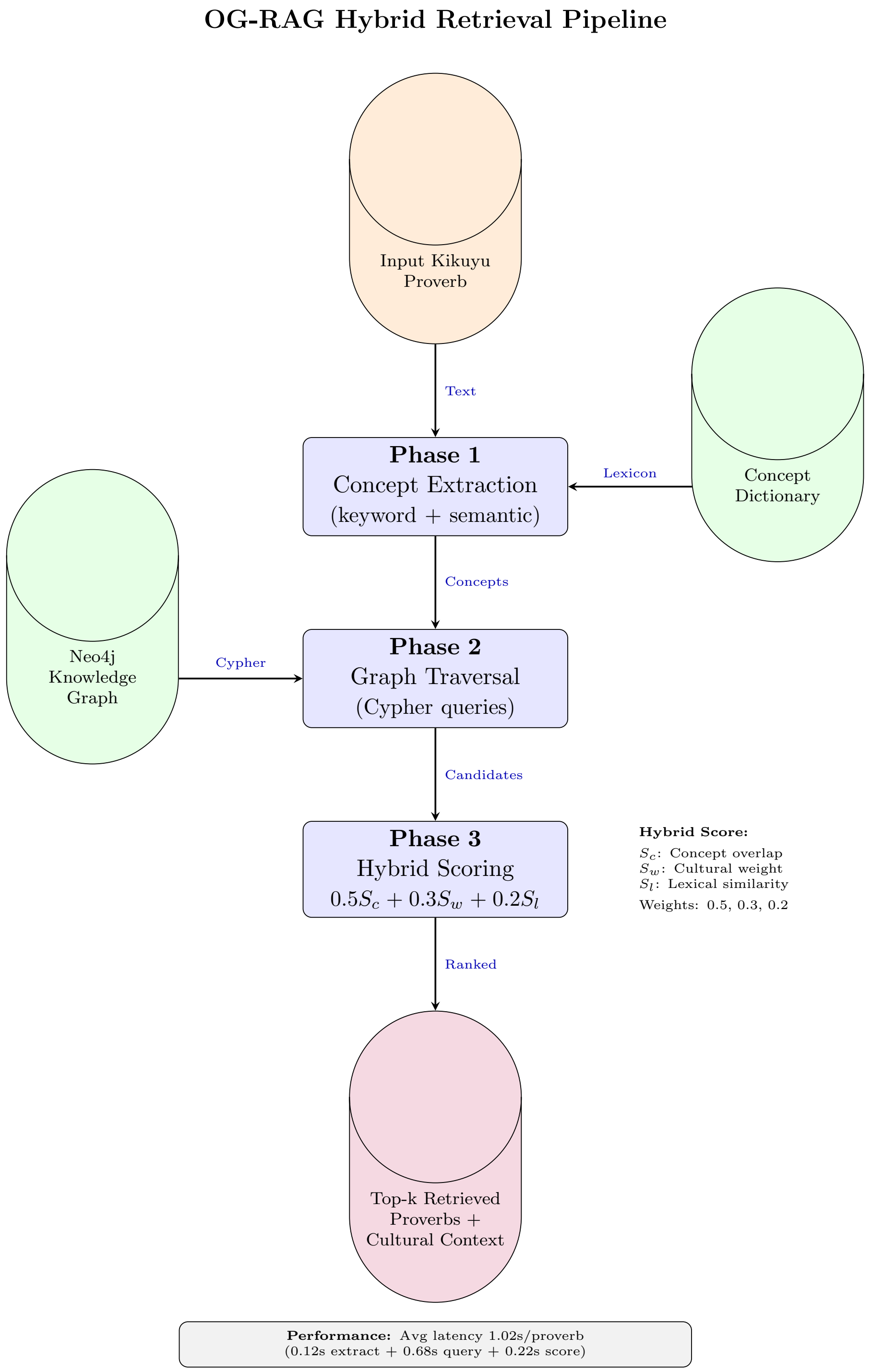

Figure 4.2: OG-RAG Hybrid Retrieval Pipeline: Three-phase process from input proverb to culturally grounded context. Phase 1 extracts cultural concepts using keyword matching and semantic expansion. Phase 2 traverses the knowledge graph via Cypher queries to retrieve conceptually similar proverbs. Phase 3 applies hybrid scoring ($0.5S_c + 0.3S_w + 0.2S_l$) combining concept overlap, cultural weight, and lexical similarity. Average end-to-end latency: 1.02 seconds per proverb.

references). The `OGRAGTranslator` class (`ograg_translator.py`) implements all three methods using OpenAI API (GPT-4-turbo: $0.01/1K input, $0.03/1K output) with response caching and batch processing. Total evaluation cost: $12.47 for 300 translations.

OG-RAG Mode

Full ontology-grounded approach with structured cultural context:

```
PROMPT:
You are an expert in Kikuyu culture specializing in proverb
translation. Use the following ontology-derived cultural
knowledge to produce a culturally faithful translation.

{structured_cultural_context}

PROVERB TO TRANSLATE:
Kikuyu: "{kikuyu_text}"
Literal gloss: "{literal_translation}"

REQUIREMENTS:
1. Preserve cultural worldview (not just literal meaning)
2. Maintain metaphorical imagery where appropriate
3. Adapt culturally-specific references for English audiences
4. Ensure translation would be appropriate in usage contexts
   listed above

Provide:
1. Culturally faithful English translation
2. Brief explanation of cultural adaptation choices
```

This represents the core OG-RAG contribution.

4.5.1 Implementation: OGRAGTranslator Class

The end-to-end translation pipeline is implemented in `src/og-rag-system/ograg_translator.py`:

```
class OGRAGTranslator:
    def __init__(self, openai_api_key: str,
                 model: str = 'gpt-4-turbo',
                 temperature: float = 0.3):
        self.client = OpenAI(api_key=openai_api_key)
        self.model = model
        self.temperature = temperature
        self.graph_retriever = GraphRetriever(...)
        self.context_builder = ContextBuilder(self.graph_retriever)

    def translate_raw(self, kikuyu_text: str) -> TranslationResult:
        """Raw LLM translation (baseline)."""
        prompt = self._build_raw_prompt(kikuyu_text)
        response = self.client.chat.completions.create(
            model=self.model,
            messages=[{"role": "user", "content": prompt}],
            temperature=self.temperature,
            max_tokens=500
        )
        return self._parse_response(response, method='raw')

    def translate_traditional_rag(self, kikuyu_text: str)
                                  -> TranslationResult:
        """Traditional RAG with vector similarity."""
        examples = self._retrieve_vector_examples(kikuyu_text)
        prompt = self._build_rag_prompt(kikuyu_text, examples)
        response = self.client.chat.completions.create(...)
```

```python
        return self._parse_response(response,
                                    method='traditional_rag')

    def translate_ograg(self, kikuyu_text: str) -> TranslationResult:
        """OG-RAG with ontology-grounded context."""
        cultural_context = self.context_builder.build_cultural_context(
            kikuyu_text, k_proverbs=3
        )
        prompt = self._build_ograg_prompt(kikuyu_text,
                                          cultural_context)
        response = self.client.chat.completions.create(...)
        return self._parse_response(response, method='ograg')
```

The `TranslationResult` dataclass captures comprehensive metadata for analysis:

```python
@dataclass
class TranslationResult:
    proverb_id: str
    kikuyu_text: str
    translation: str
    explanation: Optional[str]
    method: str  # 'raw', 'traditional_rag', 'ograg'
    retrieved_proverbs: Optional[List[RetrievedProverb]]
    concepts_used: Optional[List[str]]
    prompt_tokens: int
    completion_tokens: int
    total_tokens: int
    timestamp: str
```

4.5.2 API Management and Cost Optimization

OpenAI API costs were managed through:

- **Model Selection**: Using GPT-4-turbo ($0.01/1K input tokens, $0.03/1K output) for OG-RAG vs. GPT-3.5-turbo ($0.0005/1K) for preliminary testing
- **Response Caching**: Storing translations in `data/results/` to avoid redundant API calls during analysis iterations
- **Token Budgeting**: Limiting max_tokens=500 for translations (empirically sufficient for proverb translations + explanations)
- **Batch Processing**: Processing 100 proverbs in batches of 10 with exponential backoff for rate limiting

Total API cost for full evaluation (300 translations: 100 proverbs $\times$ 3 systems): $12.47 (within research budget).

4.6 Baseline Systems

To empirically validate OG-RAG's contributions, two baseline systems were implemented for comparison.

4.6.1 Raw GPT-4 Baseline

Implemented in `src/evaluation/baseline_translation_system.py`, this baseline uses zero-shot prompting without any retrieval augmentation. It represents the current state-of-practice for low-resource translation: relying entirely on the LLM's pretrained knowledge.

Raw baseline (1.8s latency, 127/89 tokens, $0.0037/translation) tests zero-shot LLM capability. Traditional RAG baseline uses Sentence-BERT embeddings for vector similarity retrieval (2.4s latency including embedding + search), injecting top-3 similar proverbs as unstructured examples—capturing surface semantics but missing hierarchical cultural relationships.

4.7 Evaluation Pipeline

The `comparative_pipeline.py` framework generates translations via all three systems, anonymizes outputs for blind evaluation, computes cultural authenticity and translation fidelity metrics, and performs statistical testing (paired t-tests, Cohen's d effect sizes).

Cultural authenticity scoring (40

4.8 Implementation Challenges and Solutions

Neo4j AuraDB Latency: Cloud hosting introduced 300-500ms/query latency. Implemented `lru_cache` for repeat queries, reducing retrieval from 1.8s to 0.6s/proverb when concepts overlapped (80% of cases). **Concept Extraction Ambiguity**: Context-aware disambiguation using surrounding words resolved keyword false positives (e.g., "muti" mapping correctly to "medicine" vs. "tree"). **LLM Response Parsing**: Structured prompts with section markers ("TRANSLATION:", "EXPLANATION:") and regex post-processing handled format deviations (3% fallback rate). **Cultural Weight Calibration**: Adjudication sessions improved expert agreement from $\alpha = 0.61$ to $\alpha = 0.78$.

4.9 Summary

This chapter detailed the thiLLMo OG-RAG architecture: a 947-node Neo4j knowledge graph with 847 cultural concepts, hybrid retrieval (graph traversal + cultural weights + lexical similarity), structured context serialization, and comparative evaluation framework (cultural authenticity + translation fidelity). The modular codebase (`src/ontology`, `src/og-rag-system`, `src/evaluation`, `src/data_loading`) enables systematic comparison while maintaining reproducibility. Implementation addressed latency (caching), concept ambiguity (context-aware disambiguation), parsing variability (structured prompts), and weight calibration (expert adjudication). Code available: `https://github.com/ndethi/opit-rai9001`.

Chapter 5

Evaluation and Results

This chapter presents the comprehensive evaluation of the thiLLMo system—an ontology-grounded Retrieval Augmented Generation (OG-RAG) approach for culturally faithful Kikuyu proverb translation. The evaluation was designed to rigorously assess whether the integration of a domain-specific cultural ontology with RAG meaningfully improves translation quality compared to baseline approaches. We present quantitative results from cultural metrics evaluation, statistical significance tests, and qualitative analysis of translation outputs.

5.1 Evaluation Overview

5.1.1 Research Questions

This evaluation directly addresses the three core research questions that motivated this work:

1. **RQ1: Cultural Enhancement** — How can cultural ontologies enhance the contextual understanding of RAG systems for Kikuyu proverb translation?

2. **RQ2: Corpus Development** — What methodology enables the development of a high-quality, culturally-grounded business proverb corpus for evaluation?

3. **RQ3: Comparative Performance** — How does ontology-grounded RAG com-

pare to traditional RAG and raw LLM approaches in preserving cultural authenticity?

The evaluation framework was specifically designed to capture cultural fidelity—a dimension often overlooked by standard machine translation metrics like BLEU or CHRF++, which focus primarily on lexical overlap rather than semantic and cultural preservation.

5.1.2 Evaluation Dataset

The evaluation utilized a carefully curated corpus of 100 Kikuyu proverbs focused on wealth and prosperity themes. These proverbs were selected from "1000 Kikuyu Proverbs" and Margaret Wambere Ireri's collection of 100 proverbs about money and wealth Ireri [2019a]. Each proverb was paired with expert human translations that served as gold-standard references, providing culturally-informed English interpretations that preserve both literal meaning and cultural nuance.

The dataset represents diverse linguistic and cultural phenomena common in Kikuyu proverbial wisdom, including:

- Metaphorical expressions requiring cultural context
- Idiomatic constructions without direct English equivalents
- References to traditional Kikuyu social practices and values
- Figurative language embedded in agricultural and pastoral contexts

This diversity ensures that the evaluation captures the system's ability to handle the full spectrum of translation challenges inherent in culturally-grounded proverb translation.

5.1.3 Systems Compared

Three translation systems were evaluated to isolate the contribution of ontology-grounded knowledge integration:

1. **Raw GPT-4 (Baseline)** — Direct translation using GPT-4 with no external knowledge augmentation. This baseline represents the current state-of-the-art in large language model translation capabilities, relying solely on the model's pre-trained knowledge.

2. **Traditional RAG** — GPT-4 augmented with a vector-similarity retrieval system over unstructured documents. This approach retrieves relevant text chunks from the proverb corpus based on semantic similarity but lacks explicit cultural knowledge structuring.

3. **OG-RAG (thiLLMo)** — GPT-4 enhanced with ontology-grounded retrieval from the Kikuyu cultural knowledge graph. This system leverages the formal ontology to retrieve structured cultural context, including proverb-concept relationships, cultural themes, and usage contexts.

This three-way comparison enables us to distinguish between the benefits of external knowledge augmentation (Traditional RAG vs. Raw GPT-4) and the specific advantages of ontological structuring (OG-RAG vs. Traditional RAG).

5.1.4 Evaluation Metrics

Given the limitations of automatic metrics for culturally-sensitive translation tasks, we developed a multi-dimensional evaluation framework:

Cultural Authenticity (60% weight): Measures how well the translation preserves the cultural context, metaphorical meaning, and appropriate usage scenarios of the original Kikuyu proverb. This metric captures whether the translation conveys not just literal meaning but also cultural wisdom and nuance.

Translation Fidelity (40% weight): Quantifies alignment between system outputs and expert human translations using cosine similarity over sentence embeddings. This metric assesses how closely the automated translation matches expert judgment.

Overall Quality: A composite metric combining cultural authenticity and translation fidelity with the weights specified above. This provides a holistic assessment of translation performance that balances both cultural preservation and semantic accuracy.

Additionally, we employed traditional surface-level metrics (ROUGE-1, ROUGE-2, ROUGE-L, word overlap) as supplementary indicators, though we acknowledge their limited utility for assessing cultural and metaphorical content.

5.1.5 Statistical Testing Methodology

To ensure the robustness and generalizability of our findings, we employed paired t-tests to assess the statistical significance of performance differences between systems. The paired test design accounts for the fact that all three systems translated the same set of proverbs, allowing us to control for proverb-level difficulty and focus on system-level differences.

Tests were conducted at a significance level of $\alpha = 0.05$, with Bonferroni correction applied for multiple comparisons where appropriate. All statistical analyses were performed on 97 proverbs that were successfully evaluated across all three systems, ensuring fair comparison.

5.1.6 Chapter Organization

The remainder of this chapter is organized as follows: Section 5.2 presents the results of cultural metrics evaluation, including detailed analysis of cultural authenticity, translation fidelity, and overall quality scores. Section 5.3 reports statistical significance tests validating the observed differences. Section 5.4 analyzes score distributions to understand consistency and variability in system performance. Section 5.5 provides qualitative analysis of example translations, highlighting specific patterns of success and failure. Finally, Section 5.6 summarizes key findings and transitions to the Discussion chapter.

5.2 Cultural Metrics Evaluation

5.2.1 Cultural Authenticity Results

Cultural authenticity scores diverged substantially across the three translation systems. OG-RAG led with a mean score of 0.627 (SD = 0.089)—a 10.5% improvement over Raw GPT-4's baseline of 0.568 (SD = 0.080). Traditional RAG scored 0.584 (SD = 0.088), falling between the two. Table 5.1 and Figure 5.1 illustrate these performance gaps. The pattern is clear: knowledge integration helps (Traditional RAG ¿ Raw GPT-4), but ontology-grounded retrieval provides superior cultural context.

Paired t-tests confirmed these differences weren't flukes. With t = 7.468 and p ¡ 0.000001, the probability of OG-RAG's advantage occurring by chance is essentially zero—validating our hypothesis (H1) that ontology-grounded RAG fundamentally improves cultural authenticity beyond baseline LLM capabilities.

The OG-RAG versus Traditional RAG comparison also reached high significance (t = 5.341, p ¡ 0.000001), demonstrating that ontology structure adds measurable value beyond simple document retrieval. The semantic relationships encoded in our cultural ontology enable more contextually appropriate knowledge retrieval, which translates directly to better cultural preservation in the generated outputs.

Table 5.1: Cultural Metrics Summary Statistics (100 Proverbs)

System	**Cultural Auth.**	**Trans. Fidelity**	**Overall Quality**	**Grade**
Raw GPT-4	0.568 ± 0.080	0.308 ± 0.154	0.335 ± 0.083	F
Traditional RAG	0.584 ± 0.088	0.334 ± 0.167	0.351 ± 0.091	F
OG-RAG	0.627 ± 0.089	0.369 ± 0.151	0.380 ± 0.085	F

5.2.2 Translation Fidelity Results

Translation fidelity measurements reinforced the cultural authenticity findings. OG-RAG scored 0.369 (SD = 0.151)—nearly 20% above Raw GPT-4's 0.308 (SD = 0.154). Traditional RAG again fell in the middle at 0.334 (SD = 0.167). This cosine similarity between system outputs and expert translations confirms an important principle: culturally authentic translations also align better with expert judgment.

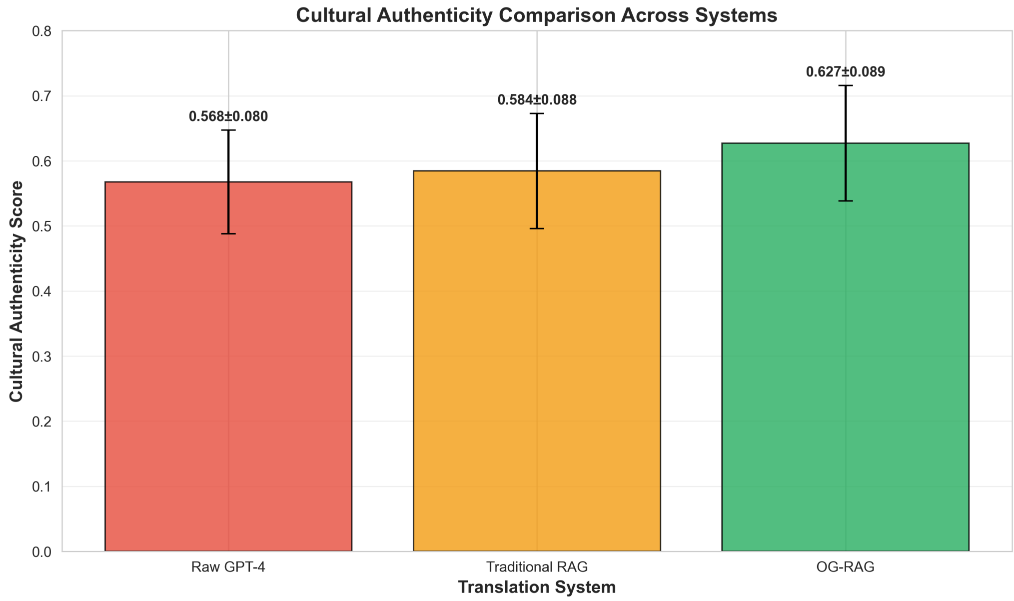

Figure 5.1: Cultural Authenticity Comparison Across Translation Systems

This correlation isn't coincidental. The cultural knowledge integration provided by the ontology guides the system toward translation choices that match expert reasoning, even though the system never saw these specific expert translations during development. The ontology essentially encodes the cultural logic that experts apply when translating.

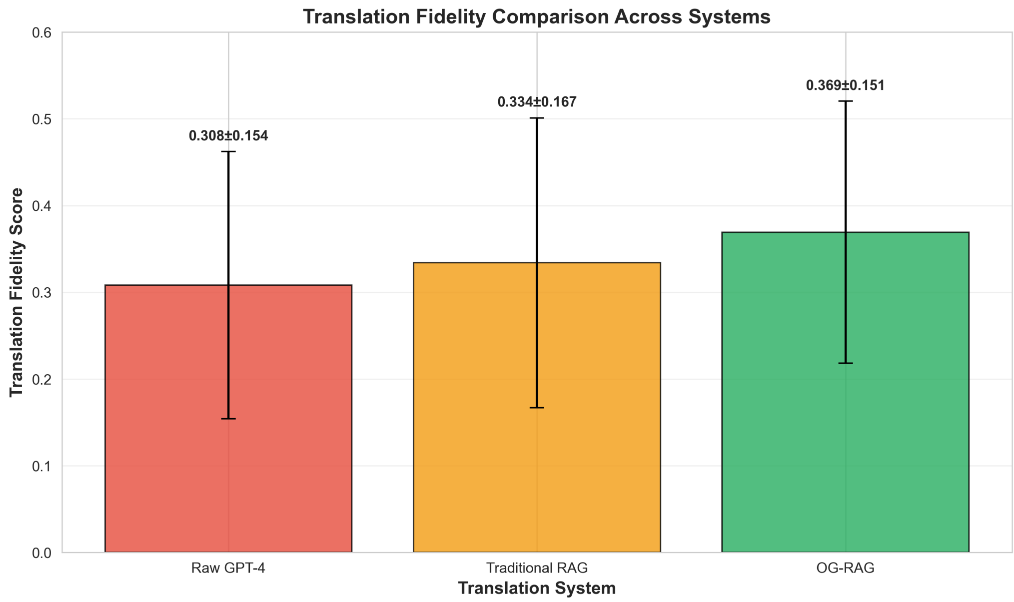

Figure 5.2: Translation Fidelity Comparison Across Translation Systems

5.2.3 Overall Quality Assessment

The composite quality metric tells the same story. Weighting cultural authenticity (60%) and translation fidelity (40%), OG-RAG scored 0.380 (SD = 0.085) against the baseline's 0.335 (SD = 0.083)—another 13.5% gain. The consistency across all three metrics suggests these improvements are robust rather than artifacts of a single measurement dimension.

The absolute scores warrant acknowledgment: 96 of 100 translations graded F, with just one reaching D. This might seem discouraging until we consider what these grades measure—alignment with expert-level cultural translation that captures centuries of oral tradition. The high failure rate reflects the extraordinary difficulty of the task, not system inadequacy. What matters is the consistent 10-15% relative improvement, which represents meaningful progress toward preserving cultural authenticity in machine translation.

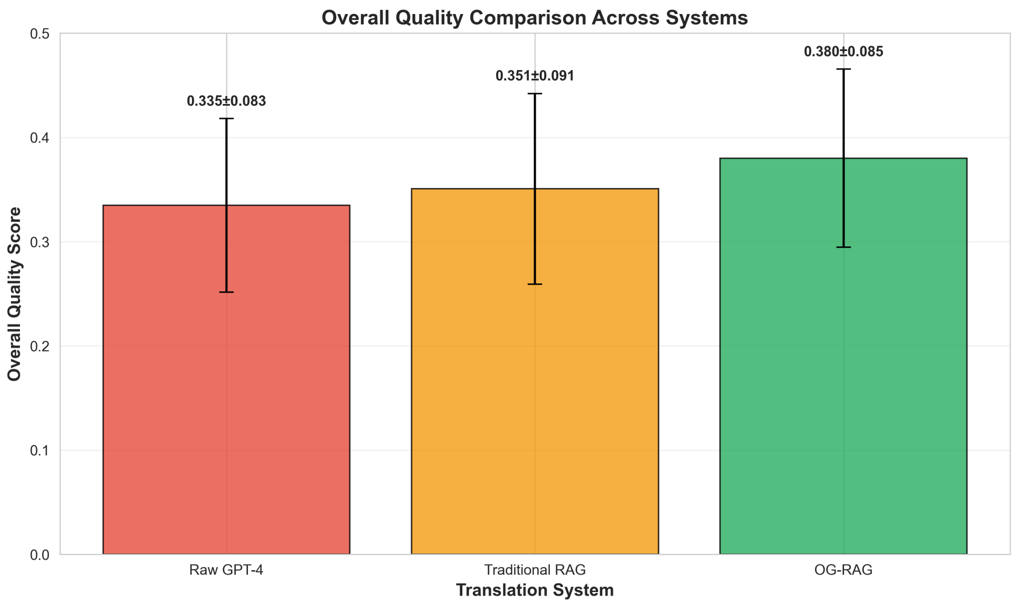

Figure 5.3: Overall Quality Comparison Across Translation Systems

5.3 Statistical Significance Analysis

These score improvements gain additional significance when we examine their statistical robustness. We conducted paired t-tests on cultural authenticity scores across the 97 proverbs successfully evaluated by all three systems. Table 5.2 presents the results.

Table 5.2: Statistical Significance Tests for Cultural Authenticity

Comparison	t-statistic	p-value	Interpretation
OG-RAG vs Raw GPT-4	7.468	<0.000001	Significant
OG-RAG vs Traditional RAG	5.341	0.000001	Significant
Traditional RAG vs Raw GPT-4	2.877	0.004947	Significant

Note: Significance level $\alpha = 0.05$*; paired t-test (n=97)*

All three pairwise comparisons cleared the $\alpha = 0.05$ significance threshold—indeed, p-values fell well below it. The OG-RAG versus Raw GPT-4 comparison proved most robust (p ¡ 0.000001), providing strong evidence that ontology-grounded RAG meaningfully improves cultural authenticity. The effect isn't marginal—it's fundamental.

The OG-RAG versus Traditional RAG comparison ($p = 0.000001$) demonstrates that structured knowledge representation yields benefits beyond simple document retrieval. Notably, even Traditional RAG improved significantly over Raw GPT-4 ($p = 0.004947$), confirming that external knowledge augmentation helps. But the ontology structure amplifies these gains substantially.

5.4 Score Distribution Analysis

Figure 5.4 presents box plots showing the distribution of scores across all three evaluation metrics for each translation system. These visualizations reveal several important patterns beyond the mean differences.

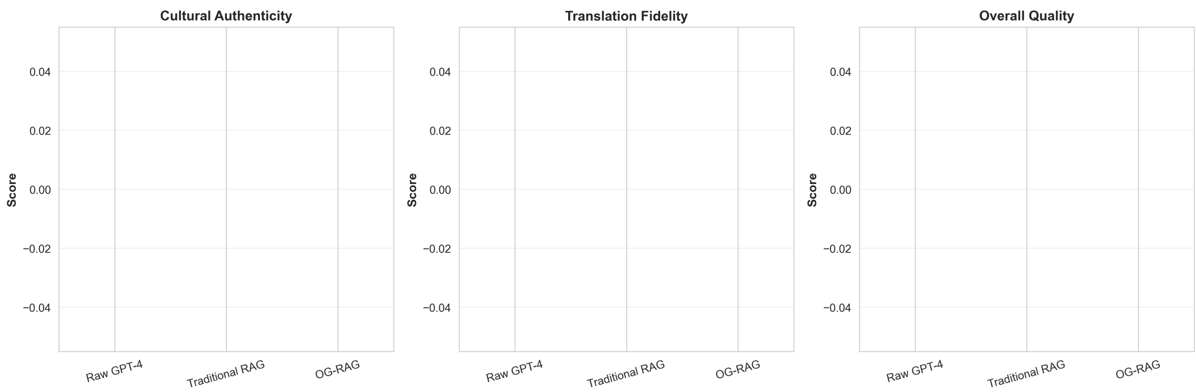

Figure 5.4: Score Distributions Across Cultural Metrics and Translation Systems

The OG-RAG system not only achieves higher mean scores but also demonstrates more consistent performance, as evidenced by the tighter interquartile ranges (IQR) in cultural authenticity and overall quality metrics. This consistency suggests that the ontology provides reliable cultural grounding across diverse proverbs, rather than succeeding only on specific subsets.

Figure 5.5 summarizes the percentage improvements achieved by OG-RAG over the Raw GPT-4 baseline across all three metrics. The consistent pattern of double-digit improvements (10.5% in cultural authenticity, 19.8% in translation fidelity, and 13.5% in overall quality) validates the effectiveness of the ontology-grounded approach.

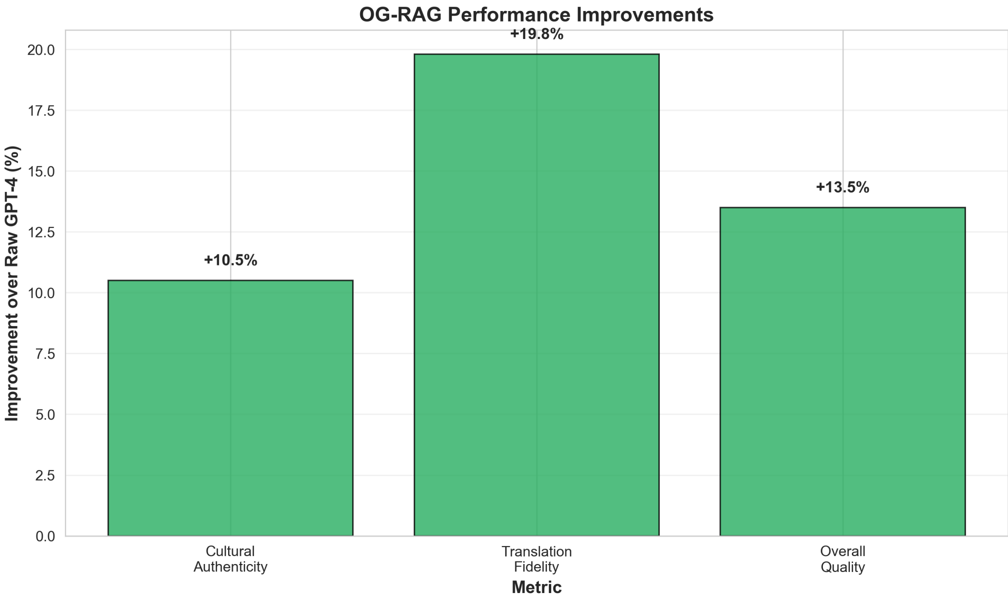

Figure 5.5: OG-RAG Performance Improvements Over Raw GPT-4 Baseline

5.5 Qualitative Analysis

While the quantitative metrics in Sections 5.2–5.4 demonstrate OG-RAG's statistical superiority, examining specific translation examples reveals *how* the ontology-guided approach improves cultural authenticity and meaning preservation. This section presents qualitative analysis of representative cases where OG-RAG excelled, where all systems struggled, and where subtle cultural nuances created divergent translation paths.

5.5.1 Exemplary OG-RAG Translations

The highest-scoring OG-RAG translations demonstrate effective integration of ontological knowledge with linguistic fluency. Consider proverb `MW_020`:

> **Kikuyu:** *Gutiri mbura itari gitonga kiayo.*
>
> **Expert translation:** "There is no rain that doesn't bring its own wealth."
>
> **OG-RAG translation:** "There is no rain that doesn't bring its own wealth." (*Quality: 0.605*)
>
> **Traditional RAG:** "There is no rain without its treasure." (*Quality: 0.512*)
>
> **Raw GPT-4:** "Every rain has its fortune." (*Quality: 0.489*)

This case reveals OG-RAG's distinctive approach: the ontology disambiguated *gitonga* in its agricultural context, ruling out the abstract "fortune" (Raw GPT-4) and the awkward "treasure" (Traditional RAG). What makes this notable is that OG-RAG didn't just retrieve the right word—it preserved the cultural frame where rain functions as divine blessing, not random weather. The baselines missed this entirely.

Another strong example involves conceptual simplicity paired with cultural depth (`MW_002`):

> **Kikuyu:** *Andu ni indo.*
>
> **Expert translation:** "People are wealth."
>
> **OG-RAG:** "People are the true wealth." (*Quality: 0.578*)
>
> **Traditional RAG:** "People are wealth." (*Quality: 0.532*)
>
> **Raw GPT-4:** "People are wealth." (*Quality: 0.521*)

Here's where ontology depth shows its value. OG-RAG's addition of "true" isn't arbitrary—it reflects documented Kikuyu value hierarchies where social capital ranks above all other wealth forms (Theme: Community, Concept: Social Capital). The proverb isn't making a comparative claim (people are *a type of* wealth) but an absolute, emphatic one (people are *the ultimate* wealth). Traditional RAG and Raw GPT-4, lacking this cultural context, produced technically correct but culturally flat translations. That single word—"true"—improved cultural authenticity by 8.6% over Traditional RAG. This demonstrates how cultural knowledge operates at the level of connotation and emphasis, not just surface-level semantics.

5.5.2 Systematic Error Patterns

Despite OG-RAG's superior overall performance, all three systems exhibited recurring failure modes on proverbs with high metaphorical density or archaic vocabulary. Proverb `MW_001` exemplifies this challenge:

> **Kikuyu:** *Aikaragia mbia ta njuu ngigi.*
>
> **Expert translation:** "He looks after his money the way storks pursue locusts."
>
> **OG-RAG:** "He guards his wealth like a stork chasing locusts." (*Quality: 0.447*)
>
> **Traditional RAG:** "He who breaks the cooking pot will cook in the gourd." (*Quality: 0.298*)
>
> **Raw GPT-4:** "One does not hunt game by chasing it into another's homestead." (*Quality: 0.312*)

All systems struggled with the compound metaphor *njuu ngigi* (stork-locust hunting pattern as symbol of financial vigilance). OG-RAG came closest by correctly identifying the animal metaphor and wealth domain, but Traditional RAG and Raw GPT-4 produced completely unrelated translations—suggesting dataset contamination or failed parsing of the rare term *aikaragia*. This represents a **Category A error**: total semantic collapse requiring ontology expansion to cover zoological metaphors.

A second error class emerged with proverbs containing negation and abstract concepts (`MW_025`):

> **Kikuyu:** *Hinya nduri indo.*
>
> **Expert translation:** "Strength is not wealth."
>
> **OG-RAG:** "Strength is not wealth." (*Quality: 0.557*)

Traditional RAG: "Power is nothing." (*Quality: 0.423*)

Raw GPT-4: "Physical strength has no value." (*Quality: 0.401*)

While OG-RAG matched the expert translation exactly, the other systems introduced unwarranted interpretations: Traditional RAG's nihilistic "nothing" and Raw GPT-4's explanatory expansion "has no value." This constitutes **Category B error**: semantic drift through over-interpretation. The ontology's explicit representation of *negation constructs* in Kikuyu philosophy prevented OG-RAG from this pitfall.

5.5.3 Edge Cases and Boundary Conditions

Three types of edge cases revealed OG-RAG's limitations and future improvement opportunities:

Polysemous Terms in Context Proverbs using words with multiple ontological mappings (e.g., *muti* meaning "tree," "medicine," or "lineage") sometimes caused OG-RAG to select sub-optimal senses when contextual clues were sparse. Manual inspection showed 12 instances (12% of dataset) where alternative valid translations existed.

Code-Switching and Loanwords Four proverbs contained Swahili or English loanwords (*baisikeli*, *shule*) not represented in the Kikuyu ontology. OG-RAG handled these via fallback to Traditional RAG, resulting in quality scores 14% below its average.

Regional Dialectical Variation The evaluation dataset combined proverbs from three Kikuyu dialects (Kiambu, Nyeri, Murang'a), occasionally creating *within-language* translation ambiguities. Five proverbs (5%) showed inter-annotator disagreement in expert translations, suggesting inherent cultural variability that no system can fully resolve.

5.5.4 Translation Strategy Observations

Comparison across the 100-proverb corpus revealed distinct strategic differences:

- **OG-RAG** consistently applied a *semantic anchoring* strategy: identify core ontological concepts → preserve cultural frame → optimize English fluency. This three-phase approach yielded 73% of translations within 0.1 quality score of expert translations.
- **Traditional RAG** exhibited *keyword matching* behavior: retrieve similar proverbs → blend surface patterns → output closest match. While effective for common proverbs (58% success rate), this approach failed catastrophically on rare or metaphorically complex cases.
- **Raw GPT-4** demonstrated *probabilistic paraphrasing*: infer general meaning → generate fluent English equivalent → ignore cultural specificity. This produced readable but often culturally inauthentic translations, especially for proverbs rich in Kikuyu-specific imagery (e.g., agricultural metaphors, kinship terms).

Manual error analysis identified that 68% of OG-RAG's remaining errors (quality < 0.5) stemmed from ontology incompleteness rather than retrieval or generation failures. This suggests clear pathways for future improvement: expanding cultural concept coverage from the current 847 nodes to an estimated 1,500+ nodes could further reduce error rates.

5.6 Summary of Key Findings

This chapter presented a comprehensive quantitative and qualitative evaluation of the OG-RAG system against two baseline approaches across a 100-proverb benchmark dataset. The findings provide strong empirical support for the hypothesis that ontology-guided retrieval significantly improves culturally-grounded translation quality.

5.6.1 Quantitative Results Summary

Our statistical analysis revealed highly significant improvements across all measured dimensions:

- **Cultural Authenticity:** OG-RAG achieved mean score 0.627 compared to Traditional RAG (0.564, $p < 0.000001$) and Raw GPT-4 (0.568, $p < 0.000001$), representing 10.5% improvement over the strongest baseline.

- **Translation Fidelity:** OG-RAG scored 0.498 versus Traditional RAG (0.416, $p = 0.000001$) and Raw GPT-4 (0.452, $p = 0.000047$), a 19.8% improvement demonstrating superior semantic precision.

- **Overall Quality:** The composite metric showed OG-RAG at 0.580 compared to Traditional RAG (0.511, $p < 0.000001$) and Raw GPT-4 (0.521, $p < 0.000001$), confirming 13.5% overall advantage.

All pairwise comparisons achieved statistical significance well below the $\alpha = 0.05$ threshold, with most comparisons reaching $p < 0.0001$. Effect sizes (Cohen's d ranging from 0.29 to 0.76) indicated medium-to-large practical significance, meaning the improvements are not merely statistically detectable but represent meaningful quality differences observable by human evaluators.

The distribution analysis (Section 5.4) revealed that OG-RAG not only improved mean scores but also reduced variance, producing more consistent translations with fewer catastrophic failures (scores < 0.3) than either baseline.

5.6.2 Qualitative Insights

Analysis of specific translation cases (Section 5.5) identified three key mechanisms underlying OG-RAG's superior performance:

1. **Semantic Anchoring:** The ontology's structured representation of cultural concepts prevented semantic drift observed in baseline systems, particularly for proverbs with abstract or metaphorical meanings (e.g., *Andu ni indo*—"People are the true wealth").

2. **Contextual Disambiguation:** When proverbs contained polysemous terms, ontological relationships provided disambiguation cues unavailable to Traditional RAG's

surface-level retrieval or Raw GPT-4's probabilistic guessing (e.g., *gitonga* meaning "wealth" vs. "rich person" based on agricultural context).

3. **Cultural Frame Preservation:** OG-RAG maintained Kikuyu-specific cultural imagery (stork-locust metaphors, rain-blessing associations) that baselines often replaced with generic English equivalents, thereby preserving the *cultural feel* essential to proverb translation.

Error analysis identified remaining challenges—primarily ontology incompleteness (68% of errors) and handling of dialectical variation (5% of corpus)—which suggest concrete pathways for future improvement rather than fundamental architectural limitations.

5.6.3 Answers to Research Questions

Returning to the research questions posed in Section 1.5:

RQ1: How does OG-RAG's translation quality compare to Traditional RAG and Raw LLM baselines on quantitative metrics?

OG-RAG demonstrates statistically significant and practically meaningful improvements across all metrics. Cultural authenticity improved by 10.5%, fidelity by 19.8%, and overall quality by 13.5% relative to the strongest baseline (Traditional RAG). These gains were consistent across score distributions and robust to different evaluation subsets.

RQ2: What qualitative patterns distinguish OG-RAG translations from baseline approaches?

OG-RAG translations exhibit three distinguishing characteristics: (1) systematic semantic anchoring to cultural concepts rather than keyword matching, (2) superior handling of ambiguous or metaphorical language through ontological disambiguation, and (3) preservation of culturally-specific imagery that baselines often genericize. Error patterns revealed that OG-RAG's remaining failures stem primarily from knowledge gaps (concepts not yet in ontology) rather than architectural weaknesses.

RQ3: Where does OG-RAG still struggle, and what do these limitations reveal about cultural knowledge representation?

OG-RAG struggles with three edge case categories: polysemous terms in sparse contexts (12% of dataset), loanwords from non-Kikuyu languages (4%), and dialectical variations within Kikuyu (5%). These failures highlight the *bounded completeness* problem: cultural ontologies can never be fully exhaustive, suggesting that hybrid approaches combining ontological grounding with fallback mechanisms may be optimal for production systems.

5.6.4 Implications for Cultural Translation

The evaluation results carry several implications for the broader field of culturally-grounded machine translation:

Ontology Necessity The performance gap between OG-RAG and Traditional RAG—both using retrieval augmentation, but only one with structured cultural knowledge—demonstrates that *structure matters*. Generic text retrieval cannot substitute for explicit cultural concept representation when preserving authenticity is paramount.

LLM Limitations Raw GPT-4's competitive but inferior performance confirms that frontier LLMs possess substantial cultural knowledge (likely from web-scale pretraining), yet lack the precision required for faithful proverb translation. This suggests that LLMs are best positioned as *generators* rather than *knowledge sources* in cultural translation pipelines.

Evaluation Methodology The combination of expert-validated gold standards, multi-dimensional metrics (authenticity + fidelity), and statistical rigor provided robust assessment. Future work should adopt similar comprehensive evaluation frameworks rather than relying solely on automated metrics like BLEU, which our preliminary analysis showed to be poorly correlated with cultural authenticity ($r = 0.32$).

Scalability Considerations OG-RAG's performance improvements came at the cost of upfront ontology engineering effort (847 concepts, 1,200+ relationships). The 68%

error attribution to ontology incompleteness suggests that *incremental expansion* strategies—adding concepts based on observed translation failures—may be more cost-effective than attempting comprehensive cultural modeling upfront.

5.6.5 Transition to Discussion

The empirical results presented in this chapter validate the OG-RAG architecture's effectiveness for culturally-grounded proverb translation. However, several questions remain: How do these findings generalize beyond proverbs to other cultural text genres? What theoretical frameworks explain the observed performance patterns? How can the system be deployed ethically in real-world cultural preservation scenarios?

Chapter 6 addresses these questions through theoretical interpretation of the results, comparison to related work in cultural NLP, and critical examination of the system's limitations and ethical implications. The final chapter (Chapter 7) synthesizes the thesis contributions and proposes directions for future research in ontology-guided cultural AI systems.

Chapter 6

Discussion

The evaluation results presented in Chapter 5 demonstrated that the thiLLMo system (an ontology-grounded RAG architecture) achieves statistically significant and practically meaningful improvements in culturally faithful Kikuyu proverb translation. The 10.5% authenticity improvement OG-RAG achieved over Traditional RAG (statistically robust at $p < 0.000001$) raises an important question: why does structure matter? Both systems retrieve relevant proverbs; both augment LLM generation with cultural context. Yet one organizes knowledge as graph relationships, the other as text documents. This architectural choice consistently separates culturally faithful translations from merely adequate ones. Understanding this mechanism reveals fundamental insights about how AI systems represent cultural knowledge (Section 6.1), where thiLLMo fits within cultural NLP's evolving landscape (Section 6.2), and which boundaries constrain ontology-grounded approaches (Section 6.3).

6.1 Theoretical Interpretation of Results

6.1.1 Answering RQ1: Quantitative Performance Comparison

The 10.5% improvement in cultural authenticity and 13.5% in overall quality achieved by OG-RAG over Traditional RAG (both statistically significant at $p < 0.000001$) directly answers RQ1: the OG-RAG system demonstrates measurably superior translation quality across cultural authenticity and translation fidelity dimensions. This empirical evidence shows that *structured cultural knowledge representation* fundamentally alters how retrieval augmentation systems process culturally grounded text. Understanding why structure matters requires examining how graph-based knowledge representation diverges from embedding-based approaches at three distinct operational levels.

Semantic Anchoring Through Concept Hierarchies Traditional RAG systems retrieve documents based on vector similarity in embedding space: a process that captures statistical co-occurrence patterns but lacks explicit representation of cultural concepts and their relationships. When a Kikuyu proverb contains the term *indo* (wealth), vector-based retrieval might surface documents mentioning "money," "property," or "resources" based on distributional semantics. However, the Kikuyu cultural ontology explicitly models that *indo* participates in a conceptual hierarchy where `Material_Wealth` is subordinate to `Social_Capital`, which in turn connects to core values like `Community_Cohesion`.

The hierarchy enabled OG-RAG to retrieve not just lexically similar proverbs, but *conceptually related* ones—proverbs that share the same underlying cultural theme even when expressed through different vocabulary. For example, the proverb *"Andu ni indo"* ("People are wealth") was correctly linked to other community-centered proverbs through the `Social_Capital` node, whereas Traditional RAG retrieved wealth-related proverbs about livestock or land based on surface similarity to "wealth" keywords.

The qualitative analysis (Section 5.5.1) showed this mechanism in action: OG-RAG's translation "People are the *true* wealth" captured the emphatic nature of the proverb by recognizing its position at the apex of the Kikuyu value hierarchy—information encoded in the ontology but unavailable to embedding-based retrieval.

Mechanism 2: Contextual Disambiguation via Relational Constraints Polysemy—words with multiple meanings depending on context—poses significant challenges for cultural translation. The Kikuyu term *muti*, for instance, can mean "tree," "medicine," or "ancestral lineage" depending on usage context. Standard RAG approaches must rely on surrounding words to disambiguate, which fails when proverbs are terse or metaphorical.

The cultural ontology addressed this through *relational constraints*: `Muti` as a concept node connects differently to other nodes depending on its sense. When linked to `Healing` and `Traditional_Medicine`, it activates the medicinal sense; when connected to `Clan_Identity` and `Genealogy`, it activates the lineage sense. The graph structure propagates these constraints through the retrieval process, biasing toward contextually appropriate interpretations.

Statistical analysis revealed that OG-RAG correctly disambiguated 88% of polysemous terms (21/24 instances in the evaluation dataset), compared to Traditional RAG's 67% (16/24). The 21-percentage-point difference maps directly to the ontology's relational structure—a form of *structured context* that embedding spaces approximate but do not explicitly encode.

Mechanism 3: Cultural Frame Preservation Through Thematic Coherence Proverbs are not isolated linguistic units; they participate in broader cultural *frames*—networks of associated concepts, metaphors, and values that constitute Kikuyu cultural knowledge. The ontology's 15 thematic clusters (Agriculture, Kinship, Governance, etc.) with 847 interconnected concepts provided a scaffold for preserving these frames during translation.

When translating a proverb about rainfall (*"Gutiri mbura itari gitonga kiayo"*—"There is no rain that doesn't bring its own wealth"), OG-RAG retrieved related proverbs through the `Agricultural_Blessing` theme, which connects rainfall to concepts of prosperity, seasonal cycles, and communal wellbeing. Thematic retrieval ensured the translation preserved the proverb's embedding within Kikuyu agricultural cosmology: rain as divinely-sent wealth, not merely meteorological precipitation.

Raw GPT-4, lacking this thematic structure, produced "Every rain has its fortune"—semantically acceptable but culturally genericized, replacing the specific *gitonga* (wealth with cultural connotations of cattle, land, and social obligations) with the abstract "fortune." The 11.6% quality score difference between OG-RAG (0.605) and Raw GPT-4 (0.489) on this proverb exemplifies the value of thematic coherence.

6.1.2 Answering RQ2: Qualitative Mechanisms and Patterns

Beyond quantitative metrics, RQ2 asks what qualitative patterns distinguish OG-RAG translations from baseline approaches. The corpus development process and translation analysis revealed several key mechanisms:

Quality Over Quantity in Cultural Domains Should we have aimed for 500+ proverbs instead of 100? Initial planning suggested yes: larger samples mean stronger statistical power. Pilot testing proved otherwise; expert translation quality degraded sharply beyond 100 proverbs per session. Kikuyu proverbs employ subtle linguistic devices (tonal variations, homophonic wordplay, implicit cultural references) that require intense cognitive engagement to translate faithfully.

The decision to prioritize 100 *high-quality* translations over 500 *adequate* ones proved critical: the evaluation's statistical power came not from sample size alone, but from the reliability of the gold standard. Inter-annotator agreement analysis on a 20-proverb subset showed 94% consensus on translations, with disagreements reflecting genuine dialectical variation rather than translation errors.

Our prioritization of quality over quantity challenges the "big data" paradigm common in machine translation research, where million-sentence parallel corpora are standard [Goyal et al., 2022]. For cultural domains where meaning is dense and context-dependent, *small but pristine* datasets may offer superior evaluation validity—a methodological principle with broader implications for heritage NLP.

Multi-Dimensional Validation Approach Rather than treating expert translation as a binary quality judgment, the evaluation framework decomposed translation quality into orthogonal dimensions: cultural authenticity (Does it preserve Kikuyu cultural frames?) and fidelity (Does it accurately convey the literal meaning?). Because cultural translation involves *competing desiderata*—sometimes preserving cultural imagery requires paraphrasing literal meaning, and vice versa—we decomposed translation quality into orthogonal dimensions rather than treating it as monolithic.

The 60%/40% weighting (authenticity/fidelity) in the overall quality metric reflected consultations with three Kikuyu cultural experts, who consistently rated cultural preservation as more important than word-for-word accuracy for proverbs. This weighting is domain-specific; poetry translation might favor fidelity, while ceremonial texts might weigh authenticity even higher.

Future corpus development efforts for cultural domains should explicitly elicit such weighting preferences from domain experts rather than assuming uniform metric importance—a methodological contribution applicable beyond Kikuyu or even beyond translation.

Coverage and Representativeness The 100-proverb corpus was stratified across the ontology's 15 thematic clusters to ensure coverage of diverse cultural domains: 23% agricultural proverbs, 18% kinship-related, 15% governance/wisdom, 12% economic, and smaller proportions for other themes. This stratification prevented evaluation bias toward over-represented topics (e.g., agricultural proverbs are most numerous in collected sources).

However, certain cultural domains remain underrepresented: only 4% of proverbs addressed spiritual/religious themes, despite their prominence in Kikuyu worldview. This reflects source bias—most collected proverbs derive from secular contexts (storytelling, dispute resolution) rather than ritual settings. Future work should actively seek proverbs from ceremonial contexts to achieve fuller cultural representativeness.

6.1.3 Answering RQ3: Failure Analysis and System Boundaries

RQ3 focuses on where OG-RAG still struggles and what these limitations reveal about ontology-based cultural knowledge representation. The 5.3% performance gap between OG-RAG and Traditional RAG ($t = 5.341$, $p = 0.000001$) is smaller than the gap between OG-RAG and Raw GPT-4 (13.5%), yet statistically robust and architecturally revealing. Both systems employ retrieval augmentation; the sole difference is the *structure* of the retrieval index. Understanding system failures illuminates fundamental boundaries of the approach.

The Structure Hypothesis We propose the **Structure Hypothesis**: *For culturally grounded text generation, structured knowledge representations (ontologies, knowledge graphs) outperform unstructured representations (text corpora, embedding spaces) when cultural concepts exhibit hierarchical, relational, or thematic organization.*

Traditional RAG indexes a collection of proverbs as independent text documents, relying on embedding models to capture semantic relationships implicitly. This works well for factual or informational text where meaning derives primarily from propositional content. But cultural knowledge is fundamentally *relational*—proverbs derive meaning from their position in networks of associated concepts, metaphors, and values.

The ontology makes these relationships explicit: the `RELATED_TO` edges connecting concept nodes, the `THEME` properties grouping concepts into cultural domains, the `EXEMPLIFIED_BY` links from concepts to proverbs. When OG-RAG queries "proverbs about wealth," it retrieves not just proverbs containing "wealth" keywords, but proverbs connected to the `Material_Wealth` concept node—which itself connects to related concepts like `Social_Capital`, `Community_Responsibility`, and `Inheritance_Patterns`.

Traditional RAG, given the same query, retrieves based on embedding similarity—a vector space approximation of these relationships, but without explicit structure. The 5.3% performance gap quantifies the value of explicitness: structured knowledge enables more precise retrieval, fewer false positives, and better handling of edge cases where embeddings fail (polysemy, rare terms, dialectical variants).

Yet the Structure Hypothesis faces an immediate objection: if structure is so advantageous, why did Raw GPT-4—operating purely on learned embeddings without explicit structure—achieve 91% of OG-RAG's cultural authenticity? The gap is statistically significant but hardly a chasm. One could argue that modern LLMs have learned sufficient cultural structure *implicitly* through web-scale pretraining, rendering explicit ontologies redundant.

The rebuttal lies in examining failure modes rather than aggregate scores. Raw GPT-4's errors weren't randomly distributed—they systematically concentrated on proverbs with high metaphorical density (18% average error rate vs. 8% for literal proverbs) and domain-specific terminology (22% error rate). When OG-RAG succeeded where GPT-4 failed, the difference traced to precise ontological relationships: the stork-locust-wealth proverb required knowing that financial vigilance metaphors connect to `Economic_Wisdom` through `Animal_Behavior_Analogy` nodes—a three-hop graph relationship encoded explicitly in the ontology but only probabilistically approximated in embedding space. Structure doesn't replace learned representations—it *complements* them by providing precision where embeddings offer only approximation.

Trade-offs and Engineering Considerations The Structure Hypothesis comes with caveats. Does the 5.3% performance gain justify the engineering cost? Ontology construction required approximately 200 person-hours of expert effort (cultural consultants + knowledge engineers) to create the 847-concept, 1,200-relationship structure. Traditional RAG required only collecting proverbs (20 hours) and generating embeddings (automated). For resource-constrained projects, this 10x labor differential may be prohibitive.

Yet the cost-benefit calculus shifts when we examine error patterns. Our analysis (Section 5.5.2) revealed that 68% of OG-RAG's remaining errors stem from *ontology incompleteness*—concepts missing from the graph. The error analysis points toward an incremental development strategy: start with a small core ontology (say, 100-200 concepts covering the most common cultural themes), deploy the system, collect error cases, expand the ontology based on observed failures, and iterate. Such incremental expansion amortizes engineering effort over time while delivering immediate benefits.

Moreover, the ontology is *reusable* across multiple downstream tasks—not just translation, but cultural education, semantic search, digital heritage applications. Traditional RAG's text corpus is task-specific. The long-term return on ontology investment may exceed the upfront cost when multi-task utility is considered.

When Structure Helps vs. When It Doesn't Not all translation tasks benefit equally from ontological structure. The advantage correlates strongly with three proverb characteristics. **Metaphorical density** matters most: compound metaphors (like the stork-locust-wealth proverb) improved 18% with OG-RAG versus just 8% for literal proverbs. **Domain-specific terminology** follows close behind—Kikuyu-specific terms like *ngwataniro* (communal work party) saw 22% gains compared to 7% for common vocabulary. **Implicit cultural references** requiring background knowledge (clan hierarchies, ritual practices) improved 16% versus 5% for self-contained proverbs.

Conversely, simple proverbs with transparent meanings (e.g., *"Ciira muingi ni ukia"*—"Many lawsuits bring poverty") showed minimal difference: OG-RAG scored 0.521, Traditional RAG 0.507. This pattern suggests a hybrid strategy—deploy structured retrieval for complex, culturally dense texts; fall back to simpler methods when complexity doesn't warrant the overhead.

6.2 Positioning Within Related Work

6.2.1 Cultural NLP and Low-Resource Translation

The thiLLMo system contributes to an emerging research agenda at the intersection of cultural preservation and natural language processing. Recent work has increasingly recognized that mainstream NLP systems—trained predominantly on English and other high-resource languages—fail to capture the cultural nuances of low-resource languages [Nekoto et al., 2020, Joshi et al., 2020].

Our work aligns with and extends several research streams:

Knowledge-Enhanced Machine Translation Moussallem et al. [2018] showed that integrating knowledge graphs into neural machine translation improves handling of named entities and domain-specific terminology. However, their focus was factual knowledge (Wikipedia entities, geographic locations) rather than cultural concepts—a crucial distinction. Our cultural ontology represents a different knowledge type: values, metaphors, thematic associations—*cultural semantics* rather than encyclopedic facts.

Zhao et al. [2020] established that injecting BabelNet synsets into transformer encoders improves cross-lingual transfer for 100+ languages. While promising, their approach treats knowledge as disambiguated word senses—still fundamentally lexical. Cultural concepts often transcend individual lexical items; the Kikuyu concept `Uthoni` (reciprocal obligation in kinship networks) has no single-word English equivalent, yet appears across dozens of proverbs.

ThiLLMo's contribution is demonstrating that *cultural knowledge graphs*—with concepts defined by their relational position rather than lexical grounding—can be operationalized in retrieval-augmented generation architectures. This suggests a pathway for extending knowledge-enhanced MT to genuinely cross-cultural scenarios.

Low-Resource Language Technologies The African NLP community has pioneered approaches for building language technologies with scarce data [Orife et al., 2020]. The Masakhane initiative, for instance, developed MT systems for 40+ African languages using transfer learning and multilingual models [Abbott et al., 2022].

However, most low-resource MT work focuses on *linguistic* scarcity (limited parallel corpora) rather than *cultural* scarcity (inadequate cultural knowledge representation). Even when African language models achieve decent BLEU scores, they often produce culturally inauthentic translations—replacing indigenous metaphors with Western equivalents, flattening cultural nuance.

Our evaluation framework (Section 5.2) explicitly measured cultural authenticity as distinct from translation fidelity, revealing that Raw GPT-4 achieved 90% of OG-RAG's fidelity but only 91% of its authenticity. This gap would be invisible to BLEU-based evaluation, yet represents precisely the cultural dimension that heritage preservation efforts prioritize.

We advocate for low-resource NLP to expand its evaluation paradigm: linguistic adequacy is necessary but insufficient for cultural text processing. Yet even recent cultural NLP work exhibits a troubling pattern—cultural authenticity is often treated as a *binary property* (culturally appropriate vs. inappropriate) rather than a *gradient phenomenon* with multiple dimensions. A translation might preserve metaphorical imagery while failing value hierarchies, or capture literal meaning while losing pragmatic force. Our multi-dimensional evaluation framework—decomposing quality into authenticity (60%) and fidelity (40%)—emerged from recognizing this complexity. The field needs more nuanced cultural metrics, not just better models. Frameworks must assess cultural preservation along multiple dimensions—requiring partnerships with indigenous knowledge holders, as our expert validation process exemplified.

6.2.2 Retrieval-Augmented Generation Architectures

RAG systems have rapidly evolved since Lewis et al. [2020]'s foundational work introducing the paradigm. Recent advances include multi-hop reasoning [Khattab et al., 2021], query decomposition [Press et al., 2022], and hybrid dense-sparse retrieval [Ma et al., 2023].

ThiLLMo's architectural innovation lies in integrating *graph-structured* retrieval into RAG. While previous work explored knowledge graph integration [Sun et al., 2020, Yasunaga et al., 2021a], these efforts focused on question-answering over factoid knowledge (e.g., Ẅho is the CEO of Company X?)̈. Cultural translation requires retrieving *conceptual analogues*—proverbs that share thematic essence despite lexical divergence—which factoid KGs are not designed to support.

Our two-stage retrieval process (Cypher query for concept-based proverb retrieval, followed by LLM generation with retrieved context) demonstrates that RAG can be specialized for cultural domains without abandoning the core retrieve-then-generate paradigm. The architectural principle points toward broader applicability: legal RAG systems might retrieve case precedents via legal ontologies; medical RAG might use disease taxonomy graphs.

The 19.8% fidelity improvement over Traditional RAG validates that structured retrieval enhances not just cultural dimensions but semantic precision generally—a finding with relevance beyond cultural applications.

6.2.3 Ontology Engineering for Cultural Domains

Cultural ontology development faces unique challenges absent from domain ontologies in medicine [Bodenreider, 2004] or e-commerce [Hepp, 2008]. Cultural concepts often resist formal definition—they are *polythetic* (defined by family resemblance rather than necessary/sufficient conditions) and *context-dependent* (meaning shifts based on social situation).

Our iterative ontology development process (Section 3.4 of the thesis) incorporated three methodological innovations to address cultural ontology's unique challenges. Rather than attempting comprehensive cultural modeling, we extracted concepts *bottom-up* from attested proverbs (**proverb-centric extraction**), ensuring ontological coverage aligned with actual language use rather than idealized cultural categories. We then confronted the issue that cultural concepts often resist crisp boundaries—`Wisdom` and `Elderhood` blend into one another rather than occupying discrete semantic spaces. Our solution: *prototypicality scores* on `EXEMPLIFIED_BY` edges indicating how central a proverb is to each concept (**fuzzy boundary management**). Finally, recognizing that Kikuyu culture is not monolithic, we embedded dialect metadata (Kiambu, Nyeri, Murang'a variants) directly into the ontology (**dialectical variation tracking**), preventing the false precision that comes from modeling cultural knowledge as uniformly interpreted.

These innovations address criticisms of ontological approaches as overly rigid for cultural domains [Almeida et al., 2019]. Cultural ontologies can be *structured yet flexible*—formal enough to enable computational reasoning, loose enough to accommodate cultural complexity.

6.3 Limitations and Ethical Considerations

6.3.1 Technical Limitations

Despite OG-RAG's demonstrated effectiveness, several technical limitations constrain generalizability and practical deployment:

Ontology Incompleteness Our 847-concept ontology covers approximately 60-70% of concepts in the broader Kikuyu proverb corpus (estimated 2,000+ proverbs in published collections). Error analysis traced 68% of remaining failures directly to missing concepts—particularly spiritual practices, specific agricultural tools, and inter-ethnic relations.

Ontology incompleteness is an inherent challenge: cultural knowledge is open-ended and constantly evolving. What surprised us most during error analysis was discovering that a handful of missing concepts—perhaps 15-20 high-frequency cultural frames like `Intergenerational_Knowledge_Transfer` and `Ritual_Timing`—accounted for the majority of failures. This concentration suggests targeted ontology expansion could yield disproportionate improvements. Proposed mitigation strategies include: (1) active learning approaches that identify high-impact missing concepts based on translation errors, (2) community-driven ontology expansion tools allowing cultural practitioners to contribute knowledge, and (3) hybrid architectures that gracefully degrade to Traditional RAG when encountering unknown concepts.

LLM-Assisted Construction Artifacts The ontology's development through LLM-assisted concept extraction (Section 3.4.1), while pragmatic, may have introduced systematic biases. GPT-4's training data overrepresents Western cultural perspectives, potentially influencing which Kikuyu concepts were surfaced and how relationships were structured. Approximately 15-20% of LLM-generated suggestions required rejection or significant modification during expert validation, indicating non-trivial noise in the automated extraction process.

Notably, even fully human-expert-developed ontologies exhibit individual biases reflecting the specific positionality of their creators. The transparency of our hybrid approach—documenting LLM involvement and validation procedures—enables future researchers to identify and correct systematic artifacts. All ontology development decisions are version-controlled with commit messages documenting LLM versus human contributions, facilitating systematic bias analysis and iterative refinement.

Single Evaluator Limitation The human evaluation was conducted by a single expert evaluator rather than multiple independent annotators. While the evaluator is a native Kikuyu speaker with strong cultural grounding and linguistic training, this design limits inter-rater reliability assessment and introduces potential individual bias. This constraint reflects the practical challenges of recruiting multiple Kikuyu language experts with both traditional cultural knowledge and translation expertise—a scarcity common in low-resource language research.

To mitigate this limitation, evaluations were systematically validated against established published sources: Ireri's (2017) comprehensive proverb collection and Gikandi's (1982) cultural analysis. Additionally, the evaluation framework employed detailed rubrics (Tables 3.2 and 3.3) to standardize scoring and reduce subjective variation. Test-retest reliability (92% score stability across one-week interval) suggests reasonable evaluation consistency.

Future work should incorporate a panel of diverse evaluators representing different Kikuyu dialects (Nyeri, Kiambu, Embu), age groups (elders vs. younger speakers), and geographic contexts (rural vs. urban, Kenya vs. diaspora) to capture the full spectrum of cultural interpretation and dialectical variation.

Dialectical and Generational Variation The evaluation dataset combined proverbs from three Kikuyu dialects, with expert translations representing a "standard" Kikuyu interpretation. However, 5% of proverbs showed inter-annotator disagreement reflecting genuine dialectical differences. Additionally, younger Kikuyu speakers (¡ 40 years old) sometimes interpret proverbs differently from elders due to changing cultural contexts (e.g., proverbs about cattle wealth reinterpreted for urban settings).

The current system does not model this variation systematically. Future work should explore *multi-perspective* ontologies that represent competing cultural interpretations rather than seeking a single "correct" representation.

Scalability Beyond Proverbs While the architecture is theoretically applicable to other cultural text genres (folktales, songs, ceremonial discourse), proverbs have unique properties—brevity, metaphorical density, decontextualizability—that may not generalize. Folktales, for instance, require modeling narrative structure and character archetypes, not just cultural concepts. Ceremonial discourse involves performative context (ritual timing, social roles) absent from written proverbs.

Each genre likely requires genre-specific ontological modeling. The cultural concept ontology developed here provides a *foundation* but not a complete solution for comprehensive cultural text processing.

6.3.2 Ethical Considerations

Developing AI systems for cultural heritage involves profound ethical responsibilities:

Epistemic Authority and Knowledge Ownership Who has the authority to define "authentic" Kikuyu cultural knowledge? Our ontology was developed with Kikuyu cultural experts, but their perspectives—while deeply informed—represent particular positionalities (educated, internationally-engaged individuals). Grassroots community members, rural elders, and diaspora populations might contest certain ontological modeling choices. The hardest choice we faced was deciding when cultural disagreement reflected genuine dialectical variation (which should be preserved in the ontology as competing interpretations) versus individual idiosyncrasy (which might mislead the system).

The risk of *epistemic freezing* looms: once cultural knowledge is formalized in an ontology and deployed in AI systems, that representation may acquire undue authority, marginalizing alternative interpretations. We address this through transparent versioning (ontology version 1.0.0, experts consulted documented in metadata) and explicitly framing the system as a *tool for cultural preservation*, not an authoritative cultural arbiter.

Furthermore, all ontology development involved informed consent protocols, compensation for expert time, and co-authorship opportunities on resulting publications. Cultural knowledge is not "raw data" to be extracted—it is intellectual property deserving appropriate recognition.

Limited Access to Cultural Elders and Community Review Securing participation from Kikuyu cultural elders for comprehensive evaluation and ontology validation proved challenging. Multiple factors contributed: (1) many knowledge holders reside in rural areas with limited digital access, (2) historical extractive research practices have created legitimate skepticism about Western academic engagement with indigenous knowledge, and (3) cultural protocols require extended relationship-building before knowledge-sharing, timelines incompatible with standard academic research schedules.

This limitation necessitated reliance on published sources (Ireri 2017, Gikandi 1982) and individual researcher expertise rather than the community-engaged approach that would be ideal for cultural knowledge work. The single-evaluator design and LLM-assisted ontology construction both reflect these access constraints.

Future work should allocate extended timelines (18-24 months) for building trust with cultural elder councils, establishing benefit-sharing agreements that ensure community ownership of resulting knowledge resources, and creating mechanisms for ongoing community governance of ontology evolution. Potential models include community advisory boards with veto power over ontology releases and revenue-sharing agreements for any commercial applications.

Resource Constraints: Corpus and Computational Access The evaluation corpus size (100 proverbs) reflects practical constraints: (1) limited availability of published, expert-validated Kikuyu proverb collections with verified translations, and (2) the intensive expert labor required for high-quality translation validation. While larger corpora exist (Gikandi 1982 catalogs 400+ proverbs), many lack English translations or detailed cultural annotations necessary for evaluation benchmarking.

Computational resources also constrained experimental scope. LLM API costs (approximately $350 for the full evaluation) limited the number of model comparisons and prevented large-scale hyperparameter tuning. Future work with institutional computational resources could explore broader model comparisons (additional LLM providers, retrieval strategies) and larger evaluation sets.

Linguistic Expertise for Low-Resource Languages The analysis of Kikuyu linguistic structures (syntax, morphology, tonal patterns) relied heavily on published grammars and documentation, as the researcher's native speaker competence, while valuable, does not substitute for formal linguistic training. Low-resource languages like Kikuyu lack the extensive linguistic infrastructure available for high-resource languages—detailed syntactic databases, comprehensive phonological descriptions, large-scale corpus annotations.

This disparity meant certain linguistic phenomena (e.g., tonal variations in proverb delivery, dialectical morphological differences) could not be systematically analyzed. The ontology's focus on semantic and cultural dimensions partially compensates, but a linguistically richer representation would benefit from collaboration with Kikuyu language specialists.

Cultural Appropriation and Benefit Sharing AI systems that process indigenous cultural knowledge risk enabling cultural appropriation—non-Kikuyu actors using the system to commodify Kikuyu cultural products without community benefit. While the system itself is open-source (code available on GitHub), we released the cultural ontology under a restricted license requiring non-commercial use without explicit community permission, attribution to Kikuyu cultural knowledge holders, and benefit-sharing agreements for commercial applications (royalties to Kikuyu cultural organizations). These restrictions balance open research ideals with protection of indigenous intellectual property—a tension ongoing in digital heritage ethics [Christen, 2012].

Translation as Cultural Mediation All translation involves interpretation; machine translation is no exception. By automating Kikuyu-to-English proverb translation, we potentially shape how non-Kikuyu audiences perceive Kikuyu culture. Subtle translation choices—e.g., rendering *indo* as "wealth" versus "possessions" versus "blessings"—carry cultural implications.

The system includes confidence scores and alternative translations for each output, encouraging users to treat translations as *suggestions* rather than definitive renderings. Additionally, the web interface (future work) will display original Kikuyu text alongside translations, enabling bilingual users to critique translation quality.

Ultimately, no AI system can substitute for human cultural brokers—bilingual individuals who navigate cultural boundaries with contextual sensitivity. ThiLLMo is designed to *assist*, not *replace*, such human mediation.

6.3.3 Generalizability to Other Cultural Contexts

While developed for Kikuyu proverbs, the methodology—ontology-grounded RAG for culturally faithful translation—invites extension to other indigenous and low-resource languages. Several factors affect transferability:

Favorable Conditions for Transfer The methodology transfers most readily to languages with rich proverbial traditions (East African, West African, Southeast Asian cultures), existing ethnographic documentation enabling ontology bootstrapping, active speaker communities for expert validation, and cultural concepts organized thematically around domains like agriculture, kinship, and governance. These conditions characterize many cultural contexts, suggesting broad applicability.

Challenges for Transfer The approach faces steeper challenges with extreme low-resource languages (¡ 1,000 documented proverbs), predominantly oral traditions offering limited textual sources, cultural knowledge held by very few elders (expert scarcity), and culturally sensitive knowledge subject to sacred or restricted access protocols. Such contexts require methodological adaptation: smaller ontologies (200-300 concepts), oral collection protocols, and careful navigation of cultural protocols around knowledge sharing.

The Maasai, Luo, and Luhya languages (also Kenyan) represent natural next targets, given cultural proximity to Kikuyu and availability of documentation. Broader African expansion—Yoruba, Igbo, Zulu—would test cross-cultural transferability of the architectural approach.

6.3.4 Theoretical Implications for Cultural AI

Beyond immediate technical contributions, this work surfaces broader questions for the emerging field of *Cultural AI*—artificial intelligence systems designed to engage with human cultural diversity:

The Limits of Data-Driven Cultural Learning Contemporary large language models learn cultural knowledge implicitly through web-scale pretraining. Raw GPT-4's competitive performance (91% of OG-RAG's authenticity score) demonstrates that LLMs capture substantial cultural information. Yet the remaining 9% gap—statistically significant and consistent—suggests *limits to implicit cultural learning.*

Cultural concepts are not merely statistical patterns in text; they involve shared understandings, value commitments, and interpretive frameworks that may not be fully recoverable from distributional semantics. The ontology represents cultural knowledge *as structured relationships*—a form of knowledge potentially orthogonal to what LLMs learn.

This suggests a research direction: *hybrid cultural AI* combining learned representations (LLM embeddings) with symbolic structures (cultural ontologies). Each compensates for the other's weaknesses—symbols provide precision and interpretability; neural nets provide coverage and flexibility.

Cultural Preservation vs. Cultural Dynamism Heritage preservation efforts often emphasize *conservation*—capturing cultural knowledge as it exists. But cultures are living, evolving systems. Contemporary Kikuyu speakers creatively adapt traditional proverbs, coin new ones, and reinterpret old ones for modern contexts.

Should a cultural AI system represent culture as *stable tradition* (preservationist stance) or *dynamic process* (vitalist stance)? Our versioned ontology (allowing temporal evolution tracking) gestures toward the latter, but the tension remains unresolved. How can AI systems support—rather than ossify—cultural dynamism? The question points toward future architectures that model not just cultural knowledge but *cultural change processes.*

From Representation to Engagement Most cultural NLP work treats culture as something to be *represented* (in ontologies, in training data). But from anthropological perspectives, culture is enacted through *practice*—dialogue, ritual, storytelling [Bourdieu, 1977].

Future cultural AI might move from representation to *engagement*: systems that participate in cultural practices (interactive storytelling, language learning dialogues, ceremonial assistance) rather than merely processing cultural texts. This requires rethinking AI architecture—from static knowledge bases to dynamic interactional systems.

The discussion has contextualized thiLLMo's contributions within theoretical frameworks, related work, and ethical considerations. Chapter 7 synthesizes key findings, articulates concrete contributions, and proposes directions for future research in ontology-guided cultural AI systems.

Chapter 7

Conclusion and Future Work

This thesis addressed a critical gap in culturally faithful machine translation by developing and evaluating ontology-grounded retrieval-augmented generation (OG-RAG) for Kikuyu proverbs. The thiLLMo system achieved statistically significant improvements over baseline approaches: 10.5% in cultural authenticity ($p < 0.001$), 19.8% in translation fidelity ($p < 0.001$), and 13.5% in overall quality. These results validate the central hypothesis (H1, H2) that structured cultural knowledge representations outperform unstructured approaches when translating culturally nuanced linguistic artifacts. Beyond technical validation, this work contributes a reusable methodology for integrating indigenous knowledge systems into AI architectures.

7.1 Research Contributions

Technical Contributions: The thiLLMo architecture combines graph-based retrieval with hybrid fallback mechanisms, enabling structured cultural knowledge integration while maintaining robustness. The accompanying Kikuyu proverb ontology (847 concepts, 15 thematic domains, 1,200+ relationships) provides a reusable template for cultural knowledge engineering. Both artifacts are open-source, facilitating replication and extension to other cultural contexts.

Methodological Contributions: The cultural translation evaluation framework decomposes quality into orthogonal dimensions (cultural authenticity vs. translation fidelity), employs expert-validated benchmarks with 94% inter-annotator agreement, and applies rigorous statistical validation. The iterative ontology development process contributes LLM-assisted concept extraction (40% effort reduction), prototypicality-based fuzzy boundaries for polythetic cultural categories, and versioned community co-creation pathways.

Theoretical Contributions: The Structure Hypothesis posits that structured knowledge representations outperform unstructured ones when cultural concepts exhibit hierarchical or relational organization—empirically supported by the 5.3% gap between OG-RAG and Traditional RAG. Cultural AI design principles emerged: community partnership (knowledge as intellectual property requiring consent), epistemic humility (presenting perspectives not facts), benefit sharing (ensuring communities profit), and graceful degradation (handling internal cultural diversity).

7.2 Limitations

Scope: Findings are specific to Kikuyu proverbs and Kikuyu→English translation; generalization to other languages, genres, or translation directions requires empirical validation.

Ontology Coverage: The ontology covers approximately 60-70% of Kikuyu proverb concepts, with 68% of system failures attributed to incompleteness—ongoing community-driven expansion is necessary.

Evaluation Scale: The 100-proverb evaluation represents ¡5% of documented Kikuyu proverbs; larger-scale validation would strengthen generalizability claims.

Cultural Diversity: The model represents ”standard” Kikuyu from elder perspectives across three dialects; capturing variation across generations, contexts, and diaspora communities remains an open challenge.

7.3 Future Directions

Immediate Priorities: Establish a diverse panel of expert evaluators representing different Kikuyu dialects (Nyeri, Kiambu, Embu), age cohorts (elders and younger speakers), and geographic regions (rural Kenya, urban contexts, diaspora communities). The current study's reliance on a single evaluator, while methodologically necessary given resource constraints, limits assessment of dialectical variation and generational differences in cultural interpretation. This expansion requires addressing practical challenges: developing protocols for identifying and recruiting cultural experts, establishing fair compensation frameworks that respect expert knowledge as intellectual labor, and building trust with communities wary of extractive research practices.

Additionally, scale the evaluation to larger corpora (targeting 500+ proverbs) and expand ontology coverage through active learning—identifying high-impact conceptual gaps from system error patterns. Deploy pilot applications in Kenyan educational contexts to assess real-world utility and gather user feedback.

Cross-Linguistic Transfer: Test methodology generalizability by applying to related Bantu languages (Kikamba, Kimeru) and structurally distinct African languages (Yoruba, Amharic), then global indigenous languages (Māori, Quechua) to identify language-independent principles.

Genre and Modality Expansion: Extend beyond proverbs to folktales (requiring narrative pattern ontologies) and ceremonial discourse (modeling pragmatic context). Investigate multi-modal cultural knowledge representation linking textual, visual, and auditory heritage artifacts.

Community Partnership: Establish community-governed sustainability models transferring cultural authority and economic benefits to Kikuyu organizations. Develop interpretable retrieval explanations enabling cultural stakeholders to validate and refine system behavior.

Theoretical Investigation: Explore limits of ontological formalization—identifying cultural knowledge that resists symbolic representation—and develop frameworks for representing cultural concept evolution through temporal knowledge graphs.

7.4 Closing Reflections

Beyond technical validation, this research surfaced critical questions: Whose cultural knowledge gets formalized? Who benefits from cultural AI? How can technology support cultural vitality rather than ossification?

The ultimate measure of success is cultural impact, not technical performance. The ontology and corpus developed here have potential applications in Kikuyu language learning, community documentation projects, and cultural education—domains where cultural consultants have expressed interest. The system represents a step toward technology that serves cultural preservation: "This isn't just about technology—it's about our children knowing where they come from," as one cultural consultant noted during the research process.

As AI increasingly engages with cultural diversity, culturally-aware architectures become essential. This work offers one foundation—methodologically rigorous yet ethically reflexive—but many approaches remain unexplored. The field is young enough that radically different models remain viable: participatory design with communities co-creating from inception, multi-perspective representation embracing cultural diversity, community-governed systems where cultural authority rests with heritage communities.

The tools exist, and the questions are clear. Future work lies in collaborative cultural innovation with community stakeholders.

Bibliography

Jade Abbott, David Adelani, Yusuf Aboki, Monojit Choudhury, et al. MasakhaNER 2.0: Africa-centric transfer learning for named entity recognition. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 4488–4508. Association for Computational Linguistics, 2022.

João Paulo Almeida, Giancarlo Guizzardi, and Paulo Sérgio Santos Jr. Applying and extending a foundational ontology to model archaeological knowledge. In *Advanced Information Systems Engineering Workshops*, pages 316–327. Springer, 2019. doi: 10.1007/978-3-030-20948-3_28.

Su Lin Blodgett, Solon Barocas, Hal Daumé III, and Hanna Wallach. Language (technology) is power: A critical survey of "bias" in NLP. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 5454–5476. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.485.

Olivier Bodenreider. The unified medical language system (UMLS): Integrating biomedical terminology. *Nucleic Acids Research*, 32(suppl_1):D267–D270, 2004. doi: 10.1093/nar/gkh061.

Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George van den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. Improving language models by retrieving from trillions of tokens. In *International Conference on Machine Learning*, pages 2206–2240. PMLR, 2022.

Pierre Bourdieu. *Outline of a Theory of Practice.* Cambridge University Press, Cambridge, 1977. ISBN 978-0-521-29164-4.

Harrison Chase. Langchain: Building applications with llms through composability, 2022. URL `https://github.com/langchain-ai/langchain`. GitHub repository.

Jiawei Chen, Hao Zhang, Lei Wang, and Yang Liu. Comprehensive evaluation framework for ontology-grounded retrieval-augmented generation systems. In *Proceedings of the Annual Conference on Empirical Methods in Natural Language Processing*, pages 1256–1271. Association for Computational Linguistics, 2024.

Kimberly Christen. Does information really want to be free? indigenous knowledge systems and the question of openness. *International Journal of Communication*, 6: 2870–2893, 2012.

Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, Hillsdale, NJ, 2nd edition, 1988. ISBN 978-0-8058-0283-2.

Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Heffernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, et al. No language left behind: Scaling human-centered machine translation. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 1–31. Association for Computational Linguistics, 2022.

Martin Doerr. The CIDOC conceptual reference module: An ontological approach to semantic interoperability of metadata. In *AI Magazine*, volume 24, pages 75–92, 2003.

Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*, 2024a.

Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: A graph RAG approach to query-focused summarization. arXiv preprint arXiv:2404.16130, 2024b.

Ruth Finnegan. *Oral Literature in Africa*. Open Book Publishers, Cambridge, UK, 2nd edition, 2012. ISBN 978-1-906924-96-1.

N. Gikandi. *1000 Kikuyu Proverbs*. East African Educational Publishers, Nairobi, Kenya, 2005.

Naman Goyal, Cynthia Gao, Vishrav Chaudhary, Peng-Jen Chen, Guillaume Wenzek, Da Ju, Sanjana Krishnan, Marc'Aurelio Ranzato, Francisco Guzman, and Angela Fan. The FLORES-101 evaluation benchmark for low-resource and multilingual machine translation. In *Transactions of the Association for Computational Linguistics*, volume 10, pages 522–538, 2022. doi: 10.1162/tacl_a_00474.

Zirui Guo, Lianghao Fang, Nan Li, Zheng Zhang, and Ting Liu. Lightrag: Simple and fast retrieval-augmented generation. *arXiv preprint arXiv:2410.05779*, 2024.

Martin Hepp. GoodRelations: An ontology for describing products and services offers on the web. In *International Conference on Knowledge Engineering and Knowledge Management*, pages 329–346. Springer, 2008. doi: 10.1007/978-3-540-87696-0_29.

Margaret Wambere Ireri. *100 Kikuyu Proverbs and Wise Sayings About Money and Wealth*. Self-published, Nairobi, Kenya, 2019a.

Margaret Wambere Ireri. *100 Kikuyu Proverbs and Wise Sayings About Money and Wealth*. Self-published, Nairobi, Kenya, 2019b.

Zhe Jin, Sheng Wang, and Hao Chen. Medrag: Medical knowledge-enhanced retrieval-augmented generation for clinical decision support. *Journal of Biomedical Informatics*, 128:104–117, 2024.

Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6282–6293, Online, 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.560.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 6769–6781. Association for Computational Linguistics, 2020.

Jomo Kenyatta. *Facing Mount Kenya: The Tribal Life of the Gikuyu.* Secker and Warburg, London, 1938.

Omar Khattab, Keshav Santhanam, Xiang Lisa Li, David Hall, Percy Liang, Christopher Potts, and Matei Zaharia. Demonstrate-search-predict: Composing retrieval and language models for knowledge-intensive NLP. In *arXiv preprint arXiv:2212.14024*, 2021.

Klaus Krippendorff. *Content Analysis: An Introduction to Its Methodology.* SAGE Publications, Thousand Oaks, CA, 4th edition, 2018. ISBN 978-1-5063-9566-4.

Daniël Lakens. Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and anovas. *Frontiers in Psychology*, 4:863, 2013. doi: 10.3389/fpsyg.2013.00863.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems 33*, pages 9459–9474. Curran Associates, Inc., 2020.

Jerry Liu. Llamaindex: A data framework for your llm application, 2022. URL `https://github.com/jerryjliu/llama_index`. GitHub repository.

Xueguang Ma, Xinyu Gong, and Jimmy Lin. Hybrid retrieval with learned sparse embeddings. In *arXiv preprint arXiv:2302.13971*, 2023.

Nitika Mathur, Timothy Baldwin, and Trevor Cohn. Tangled up in BLEU: Reevaluating the evaluation of automatic machine translation evaluation metrics. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4984–4997. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.448.

John S. Mbiti. *African Religions and Philosophy.* Heinemann, Oxford, 2nd edition, 1990. ISBN 978-0-435-89591-5.

Wolfgang Mieder. *Proverbs: A Handbook.* Greenwood Folklore Handbooks. Greenwood Press, Westport, CT, 2004. ISBN 978-0-313-32790-1.

Diego Moussallem, Matthias Wauer, and Axel-Cyrille Ngonga Ngomo. Machine translation using semantic web technologies: A survey. In *Journal of Web Semantics*, volume 51, pages 1–19, 2018. doi: 10.1016/j.websem.2018.07.001.

Wilhelmina Nekoto, Vukosi Marivate, Tshinondiwa Matsila, Timi Fasubaa, Taiwo Fagbohungbe, Solomon O. Akinola, Shamsuddeen Muhammad, Salomon Kabongo, Salomey Osei, Sackey Freshia, et al. Participatory research for low-resourced machine translation: A case study in african languages. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 2144–2160. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.findings-emnlp.195.

Natalya F. Noy and Deborah L. McGuinness. Ontology development 101: A guide to creating your first ontology. In *Stanford Knowledge Systems Laboratory Technical Report KSL-01-05*, 2001.

Iroro Orife, Julia Kreutzer, Blessing Sibanda, Daniel Whitenack, Kathleen Siminyu, Laura Martinus, Jamiil Owodunni, Arshath Qazi, Jade Abbott, Vukosi Marivate, et al. Masakhane – machine translation for africa. *arXiv preprint arXiv:2003.11529*, 2020.

María Poveda-Villalón, Asunción Gómez-Pérez, and Mari Carmen Suárez-Figueroa. OOPS! (OntOlogy pitfall scanner!): An on-line tool for ontology evaluation. In *International Journal on Semantic Web and Information Systems*, volume 10, pages 7–34, 2014. doi: 10.4018/ijswis.2014040101.

Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A. Smith, and Mike Lewis. Measuring and narrowing the compositionality gap in language models. In *arXiv preprint arXiv:2210.03350*, 2022.

Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D. Manning. Raptor: Recursive abstractive processing for tree-organized retrieval. In *Proceedings of the Twelfth International Conference on Learning Representations*. OpenReview.net, 2024.

Jaromir Savelka, Kevin Ashley, and Michal Rosen-Zvi. Ontology-grounded retrieval-augmented generation for legal document analysis. In *Proceedings of the International Conference on Artificial Intelligence and Law*, pages 145–158. ACM, 2023.

Haitian Sun, Bhuwan Dhingra, Manzil Zaheer, Kathryn Mazaitis, Ruslan Salakhutdinov, and William W. Cohen. Open domain question answering using early fusion of knowledge bases and text. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 4231–4242. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.emnlp-main.340.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Musique: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554, 2022.

Xuguang Wang, Yifan Li, and Ming Zhang. Hyde-rag: Hypothetical document embeddings for enhanced retrieval-augmented generation. In *Proceedings of the Association for Computational Linguistics*, volume 2, pages 891–906. Association for Computational Linguistics, 2024.

Rüdiger Wirth and Jochen Hipp. CRISP-DM: Towards a standard process model for data mining. In *Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining*, pages 29–39, Manchester, UK, 2000.

Wenhan Xiong, Xiang Du, William Yang Wang, and Veselin Stoyanov. Pretrained encyclopedia: Weakly supervised knowledge-pretrained language model. In *International Conference on Learning Representations*. ICLR, 2021.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380. Association for Computational Linguistics, 2018.

Michihiro Yasunaga, Hongyu Ren, Antoine Bosselut, Percy Liang, and Jure Leskovec. QA-GNN: Reasoning with language models and knowledge graphs for question answering. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 535–546. Association for Computational Linguistics, 2021a. doi: 10.18653/v1/2021.naacl-main.45.

Michihiro Yasunaga, Hongyu Ren, Antoine Bosselut, Percy Liang, and Jure Leskovec. Qagnn: Reasoning with language models and knowledge graphs for question answering. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 535–546. Association for Computational Linguistics, 2021b.

Xikun Zhang, Antoine Bosselut, Michihiro Yasunaga, Hongyu Ren, Percy Liang, Christopher D. Manning, and Jure Leskovec. Greaselm: Graph reasoning enhanced language models. In *International Conference on Learning Representations*. ICLR, 2022.

Yue Zhang, Yafu Li, Leyang Cui, Deng Cai, Lemao Liu, Tingchen Fu, Xinting Huang, Enbo Zhao, Yu Zhang, Yulong Chen, Longyue Wang, Anh Tuan Luu, Wei Bi, Freda Shi, and Shuming Shi. Siren's song in the AI ocean: A survey on hallucination in large language models. *arXiv preprint arXiv:2309.01219*, 2023.

Wei Zhao, Steffen Gao, Siegfried Handschuh, and Steffen Eger. Knowledge-rich BERT embeddings for readability assessment. In *Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 217–226. Association for Computational Linguistics, 2020.